In [1]:
```
import pandas as pd
df=pd.read_csv("C:/Users/DELL/Downloads/city_day.csv")
```

In [3]:
```
df.shape
```

Out[3]:
```
(29531, 16)
```

In [4]:
```
df.head()
```

Out[4]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzen
0	Ahmedabad	2015-01-01			0.92	18.22	17.15		0.92	27.64	133.36	0.
1	Ahmedabad	2015-01-02			0.97	15.69	16.46		0.97	24.55	34.06	3.6
2	Ahmedabad	2015-01-03			17.4	19.3	29.7		17.4	29.07	30.7	6.
3	Ahmedabad	2015-01-04			1.7	18.48	17.97		1.7	18.59	36.08	4.4
4	Ahmedabad	2015-01-05			22.1	21.42	37.76		22.1	39.33	39.31	7.0

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype 
---  ------              --------------  ----- 
 0   City                29531 non-null  object
 1    Date               29531 non-null  object
 2    PM2.5              29531 non-null  object
 3    PM10               29531 non-null  object
 4    NO                 29531 non-null  object
 5    NO2                29531 non-null  object
 6    NOx                29531 non-null  object
 7    NH3                29531 non-null  object
 8    CO                 29531 non-null  object
 9    SO2                29531 non-null  object
 10   O3                 29531 non-null  object
 11   Benzene            29531 non-null  object
 12   Toluene            29531 non-null  object
 13   Xylene             29531 non-null  object
 14   AQI                29531 non-null  object
 15   AQI_Bucket         29531 non-null  object
dtypes: object(16)
memory usage: 3.6+ MB
```

In [6]:
```
df.tail()
```

Out[6]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3
29526	Visakhapatnam	2020-06-27	15.02	50.94	7.68	25.06	19.54	12.47	0.47	8.55	23.3
29527	Visakhapatnam	2020-06-28	24.38	74.09	3.42	26.06	16.53	11.99	0.52	12.72	30.14
29528	Visakhapatnam	2020-06-29	22.91	65.73	3.45	29.53	18.33	10.71	0.48	8.42	30.96
29529	Visakhapatnam	2020-06-30	16.64	49.97	4.05	29.26	18.8	10.03	0.52	9.84	28.3
29530	Visakhapatnam	2020-07-01	15.0	66.0	0.4	26.85	14.05	5.2	0.59	2.1	17.05

In [7]:
```
df.describe()
```

Out[7]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	
count	29531	29531	29531	29531	29531	29531	29531	29531	29531	29531	29
unique	26	2009	11717	12572	5777	7405	8157	5923	1780	4762	7
top	Ahmedabad	2020-06-26							0.0		
freq	2009	26	4598	11140	3582	3585	4185	10328	2328	3854	4

In [8]:
```
print(df.isnull().sum())
```

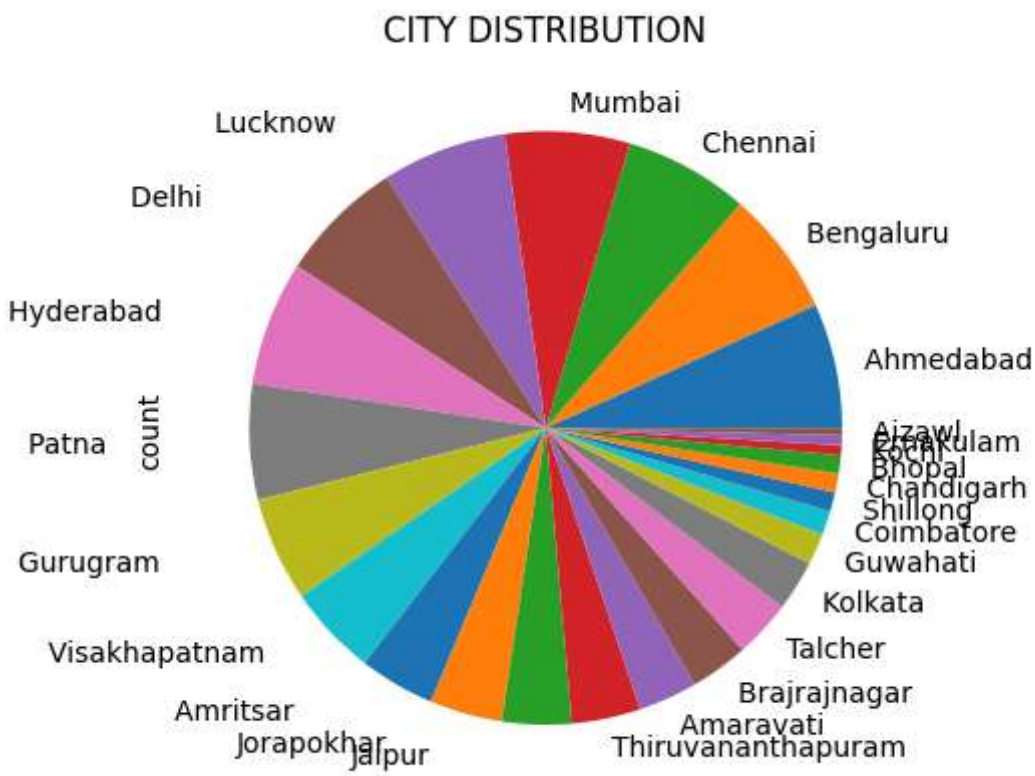
```
City                 0
 Date                0
 PM2.5               0
 PM10                0
 NO                  0
 NO2                 0
 NOx                 0
 NH3                 0
 CO                  0
 SO2                 0
 O3                  0
 Benzene             0
 Toluene             0
 Xylene              0
 AQI                 0
 AQI_Bucket          0
dtype: int64
```

In [9]:
```
print(df.dtypes)
```

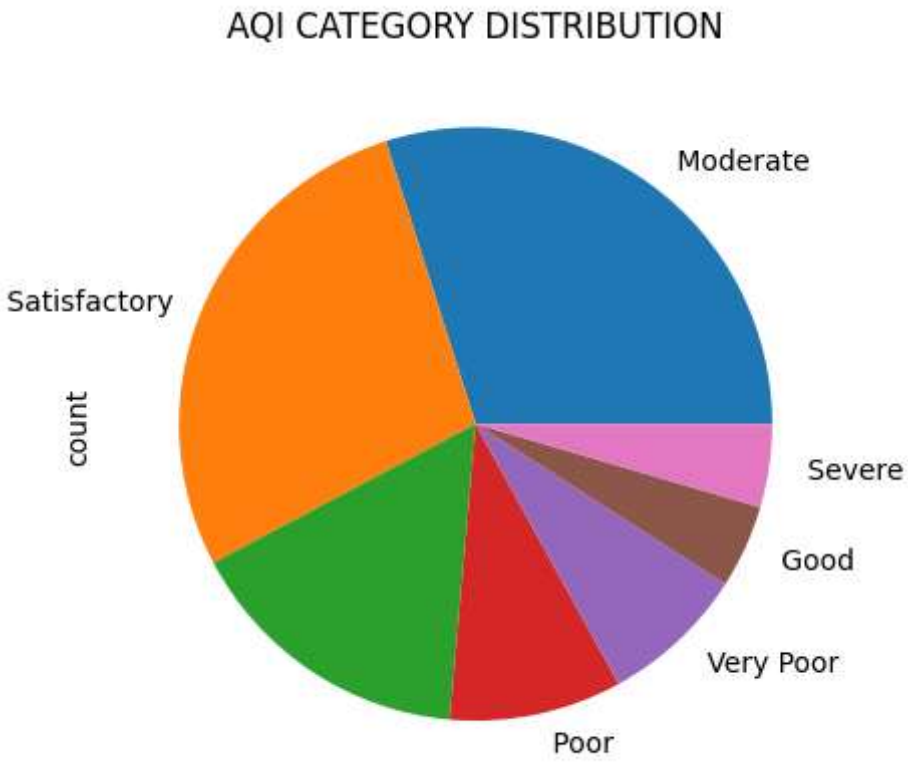
```
City                  object
 Date                 object
 PM2.5                object
 PM10                 object
 NO                   object
 NO2                  object
 NOx                  object
 NH3                  object
 CO                   object
 SO2                  object
 O3                   object
 Benzene              object
 Toluene              object
 Xylene               object
 AQI                  object
 AQI_Bucket           object
dtype: object
```

In [10]:
```
df.columns=df.columns.str.strip()
```

In [11]:
```
import matplotlib.pyplot as plt
df["City"].value_counts().plot(kind='pie',color="pink")
plt.title("CITY DISTRIBUTION")
plt.show()
```



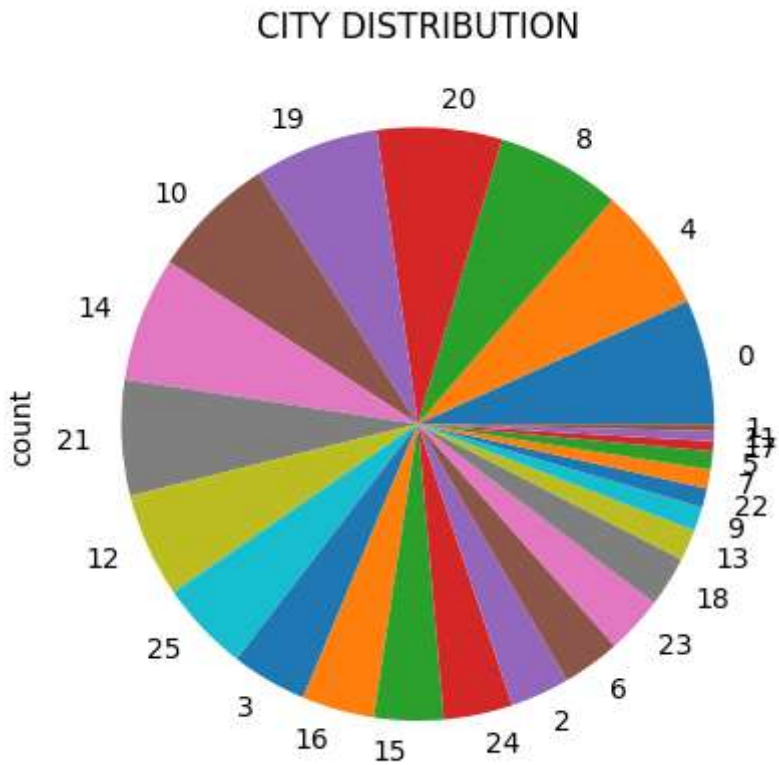
In [13]:
```
df["AQI_Bucket"].value_counts().plot(kind='pie')
plt.title("AQI CATEGORY DISTRIBUTION")
plt.show()
```



In [14]:
```
df = df.drop(['Date'], axis=1)
```

In [15]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['City'] = le.fit_transform(df['City'])
```

In [17]:
```
import matplotlib.pyplot as plt
df["City"].value_counts().plot(kind='pie',color="pink")
plt.title("CITY DISTRIBUTION")
plt.show()
```



In [18]:
```
print(le.classes_)
```

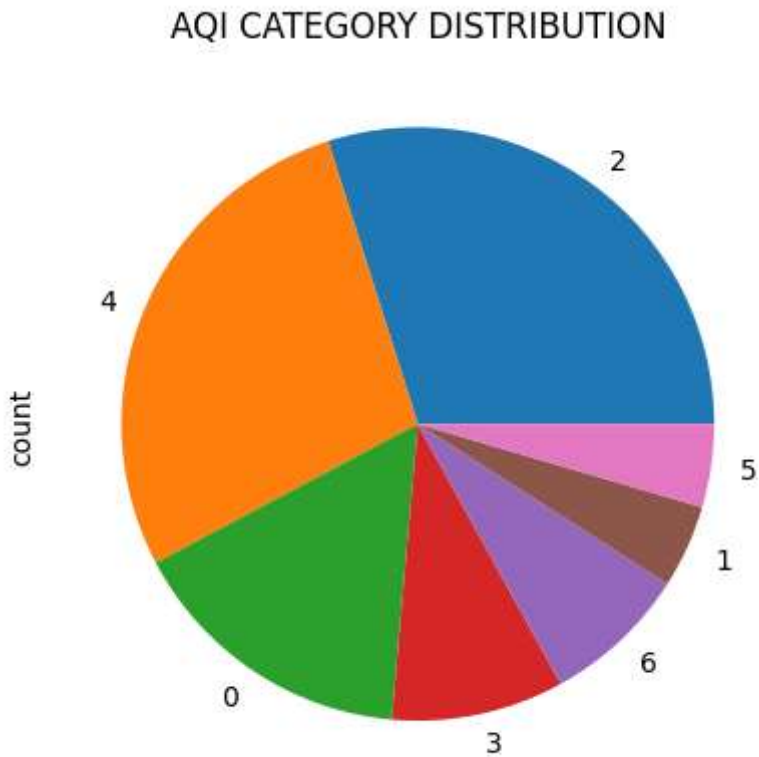
```
['Ahmedabad         ' 'Aizawl           ' 'Amaravati        '
 'Amritsar          ' 'Bengaluru        ' 'Bhopal           '
 'Brajrajnagar      ' 'Chandigarh       ' 'Chennai          '
 'Coimbatore        ' 'Delhi            ' 'Ernakulam        '
 'Gurugram          ' 'Guwahati         ' 'Hyderabad        '
 'Jaipur            ' 'Jorapokhar       ' 'Kochi            '
 'Kolkata           ' 'Lucknow          ' 'Mumbai           '
 'Patna             ' 'Shillong         ' 'Talcher          '
 'Thiruvananthapuram' 'Visakhapatnam    ']
```

In [19]:
```
from sklearn.preprocessing import LabelEncoder
le1 = LabelEncoder()
df['AQI_Bucket'] = le1.fit_transform(df['AQI_Bucket'])
```

In [20]:
```
print(le1.classes_)
```

```
[' ' ' Good' ' Moderate' ' Poor' ' Satisfactory' ' Severe' ' Very Poor']
```

In [23]:
```
df["AQI_Bucket"].value_counts().plot(kind='pie')
plt.title("AQI CATEGORY DISTRIBUTION")
plt.show()
```



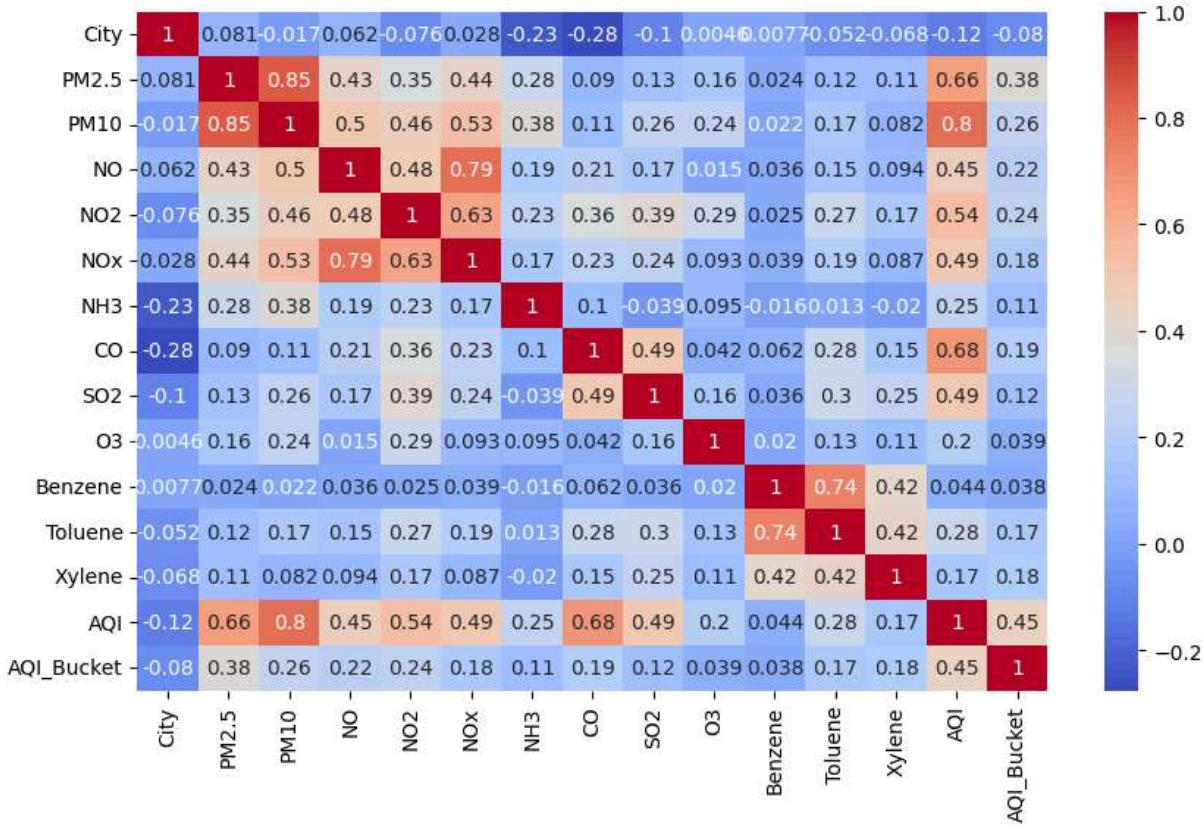
In [24]:
```
df = df.apply(pd.to_numeric, errors='coerce')
```

In [25]:
```
print(df.dtypes)
```

```
City            int64
PM2.5         float64
PM10          float64
NO            float64
NO2           float64
NOx           float64
NH3           float64
CO            float64
SO2           float64
O3            float64
Benzene       float64
Toluene       float64
Xylene        float64
AQI           float64
AQI_Bucket      int64
dtype: object
```

In [22]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

In [26]:
```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



In [27]:
```
df=df.drop_duplicates()
```

In [28]:
```
df.shape
```

Out[28]:
```
(27234, 15)
```

In [31]:
```
df["AQI_Bucket"].value_counts()
```

Out[31]:
```
AQI_Bucket
2    8829
4    8224
3    2781
0    2384
6    2337
1    1341
5    1338
Name: count, dtype: int64
```

In [32]:
```
df["City"].value_counts()
```

Out[32]:
```
City
4     2009
10    2009
19    1991
14    1986
8     1982
21    1667
12    1621
20    1484
0     1415
25    1256
3     1196
15    1106
24    1099
16     929
2      898
23     814
18     814
6      793
13     502
9      385
7      304
5      281
22     258
17     162
11     160
1      113
Name: count, dtype: int64
```

In [ ]:
```python
import numpy as np
for col in df.columns:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    print(col, "Outliers:", len(outliers))
```

```
City Outliers: 0
PM2.5 Outliers: 1982
PM10 Outliers: 1057
NO Outliers: 2459
NO2 Outliers: 1188
NOx Outliers: 1852
NH3 Outliers: 1015
CO Outliers: 2493
SO2 Outliers: 2578
O3 Outliers: 713
Benzene Outliers: 1613
Toluene Outliers: 2356
Xylene Outliers: 1005
AQI Outliers: 1358
AQI_Bucket Outliers: 0
```

In [49]:
```
for col in df.columns:
    if col != 'AQI_Bucket':
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR
        df = df[(df[col] >= lower) & (df[col] <= upper)]
```

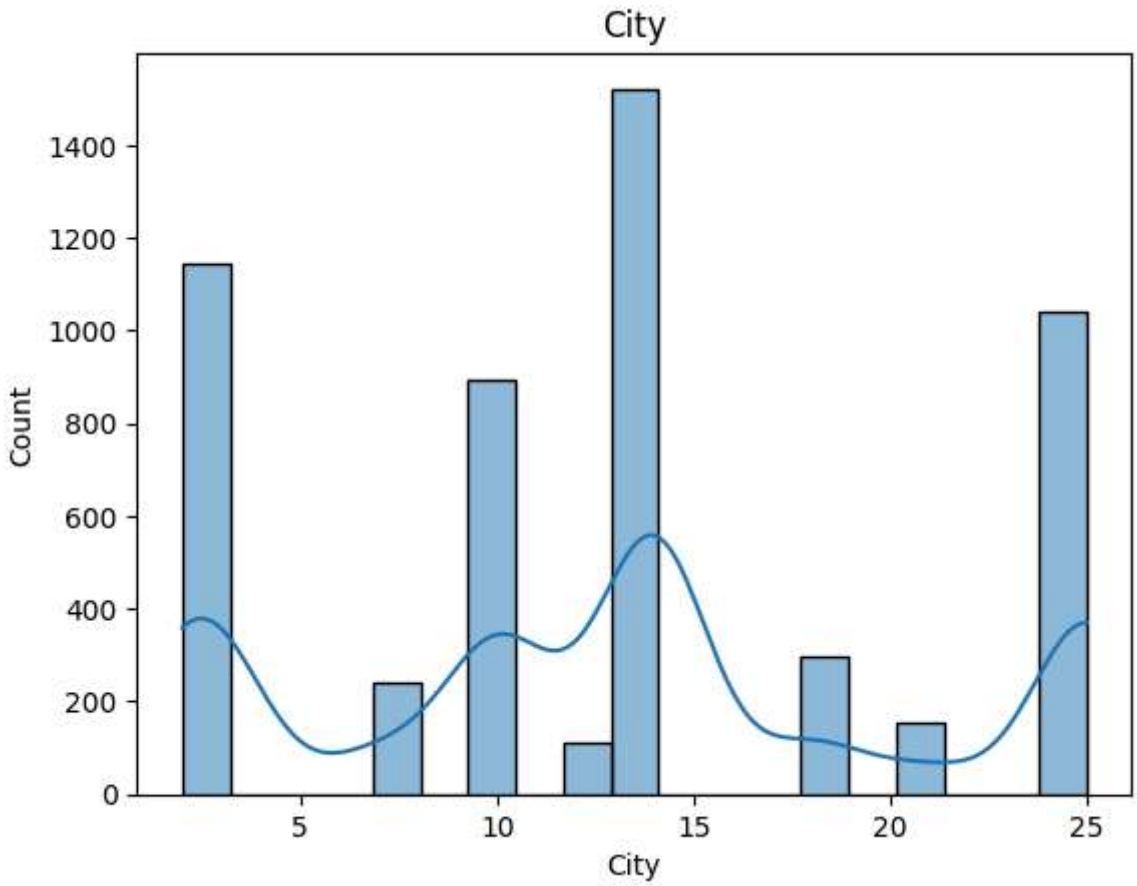
In [50]:
```
import numpy as np
for col in df.columns:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    print(col, "Outliers:", len(outliers))
```

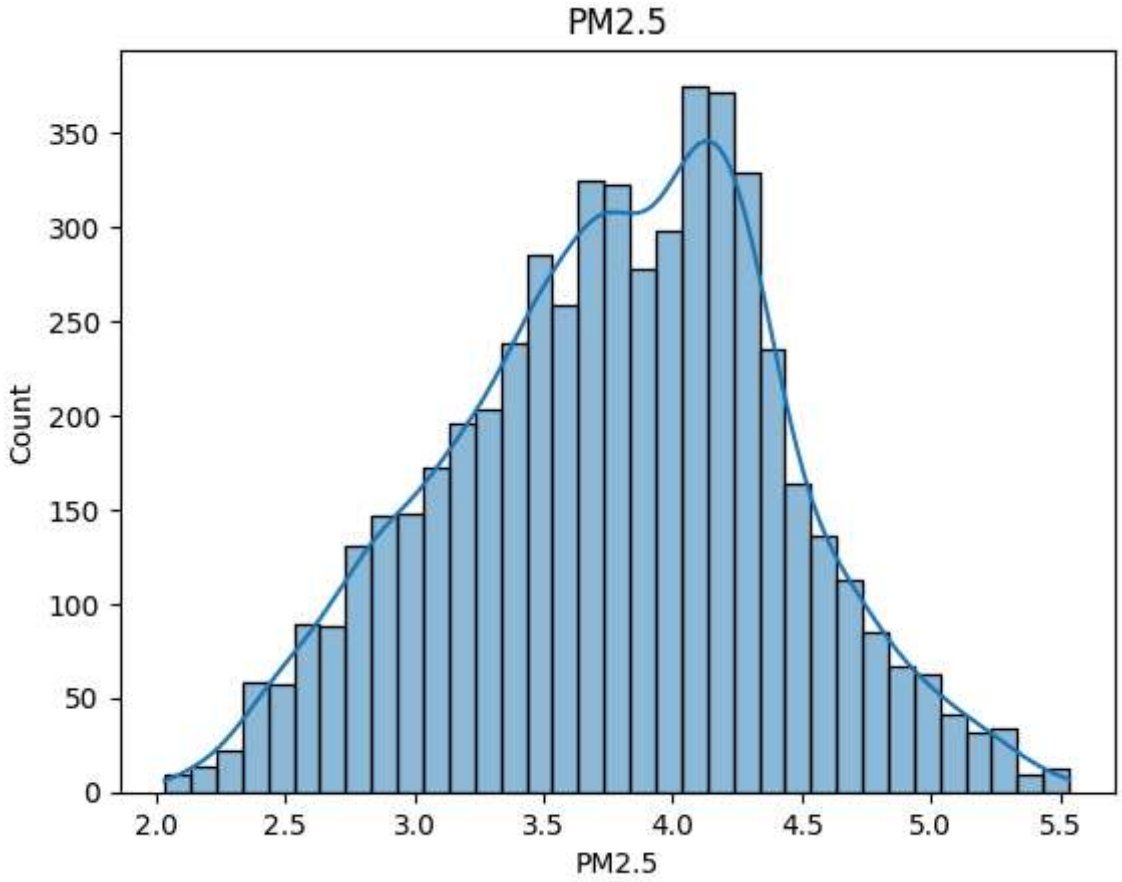
```
City Outliers: 0
PM2.5 Outliers: 0
PM10 Outliers: 0
NO Outliers: 0
NO2 Outliers: 0
NOx Outliers: 0
NH3 Outliers: 0
CO Outliers: 0
SO2 Outliers: 0
O3 Outliers: 0
Benzene Outliers: 0
Toluene Outliers: 0
Xylene Outliers: 0
AQI Outliers: 0
AQI_Bucket Outliers: 0
```

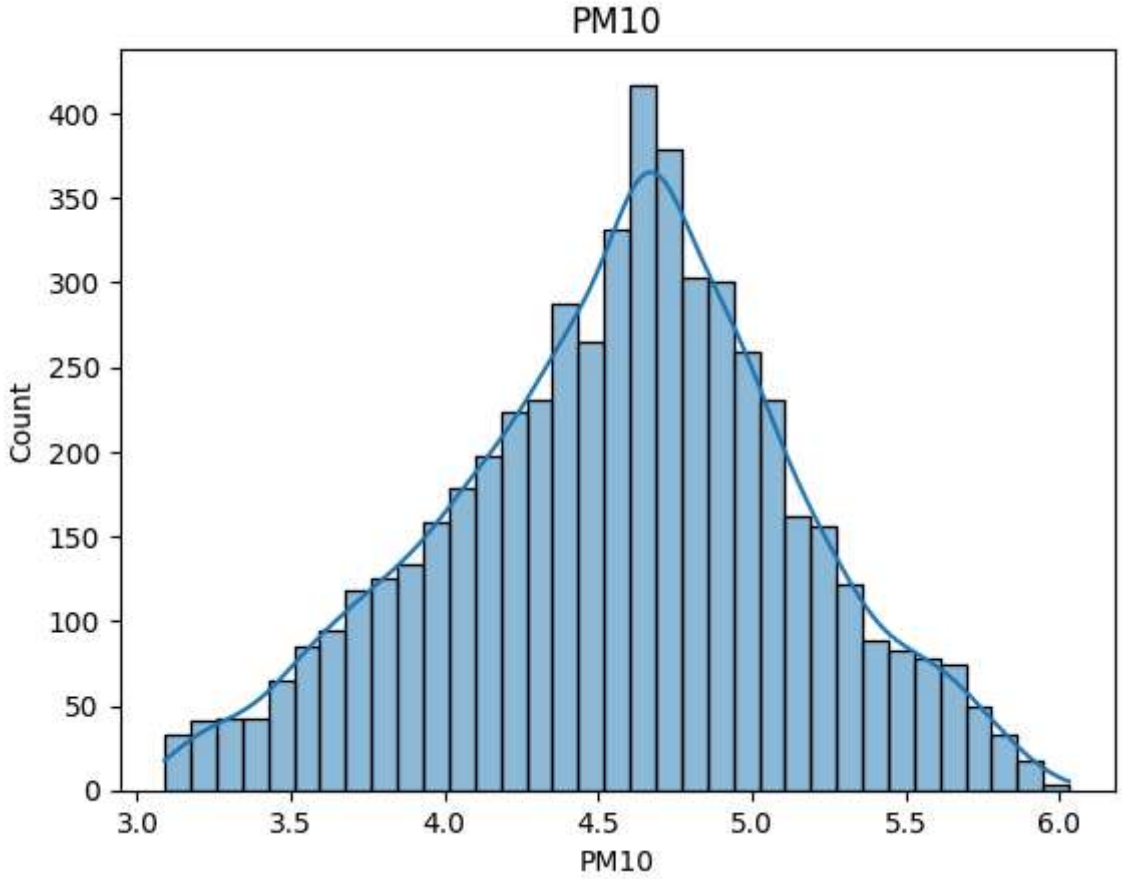
In [51]:
```
for col in df.columns:
    skew = df[col].skew()
    print(col, "Skewness:", skew)
```

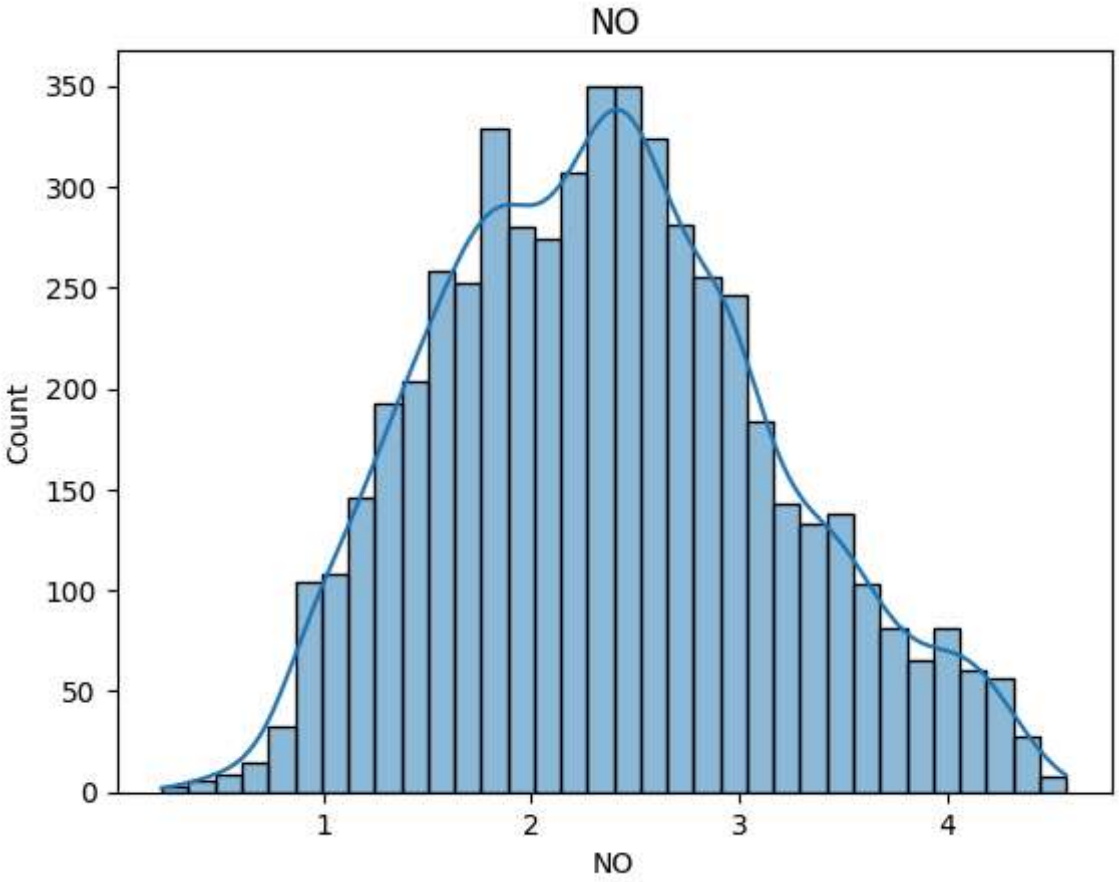
```
City Skewness: 0.1750047024234094
PM2.5 Skewness: -0.10912557967665915
PM10 Skewness: -0.18243078575478797
NO Skewness: 0.27340658289787845
NO2 Skewness: -0.3502569170156891
NOx Skewness: 0.03100601110652923
NH3 Skewness: 0.08324162490760162
CO Skewness: -0.04891811454418102
SO2 Skewness: 0.07135142960347533
O3 Skewness: -0.18722393900441092
Benzene Skewness: 0.26165373430469613
Toluene Skewness: -0.005989847833102925
Xylene Skewness: 0.7014813474556878
AQI Skewness: 0.19880961764378555
AQI_Bucket Skewness: 0.5007546881192304
```

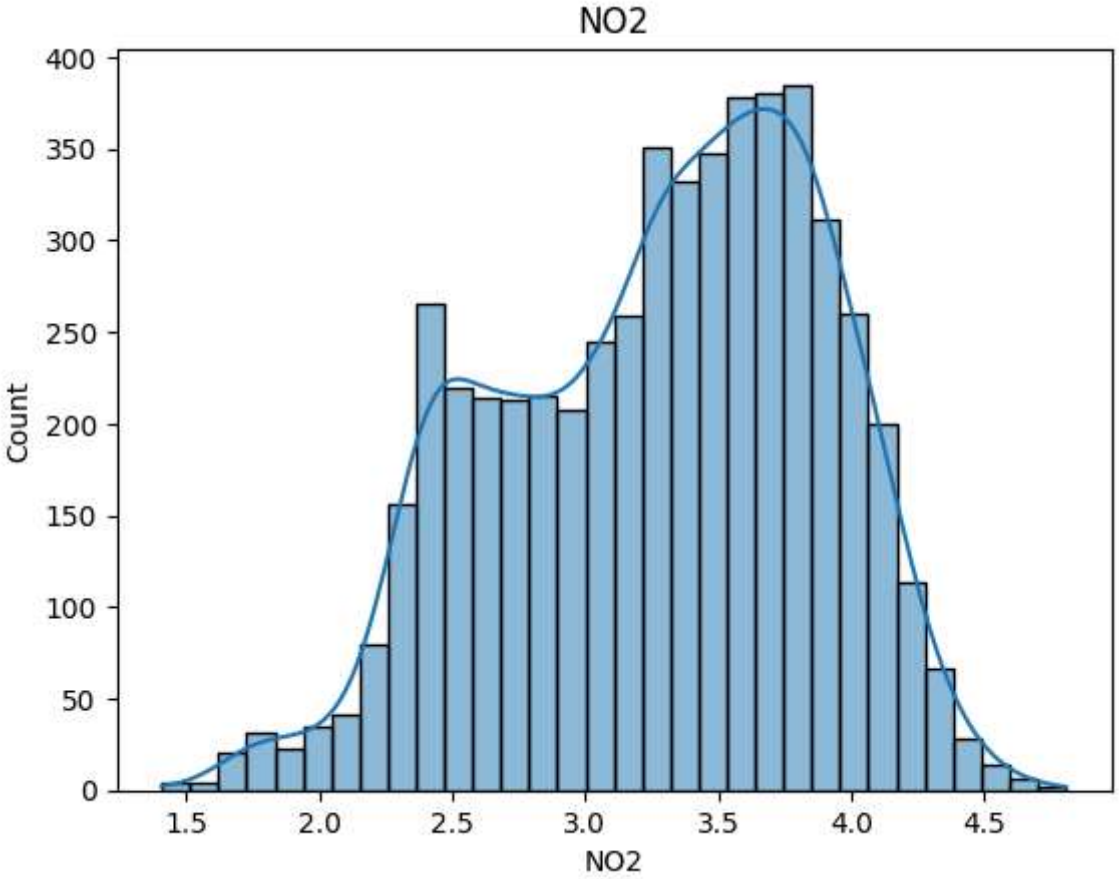
In [52]:
```
for col in df.columns:
    plt.figure()
    sns.histplot(df[col], kde=True)
    plt.title(col)
    plt.show()
```

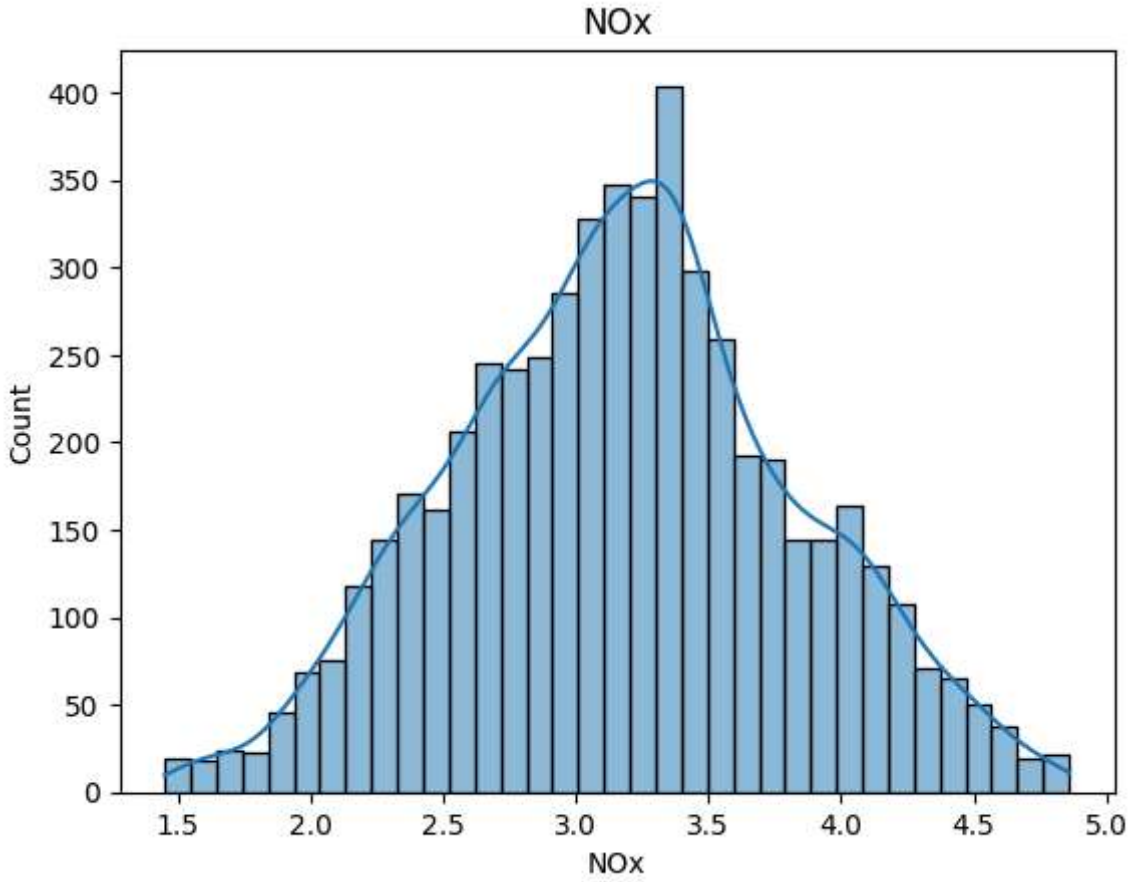


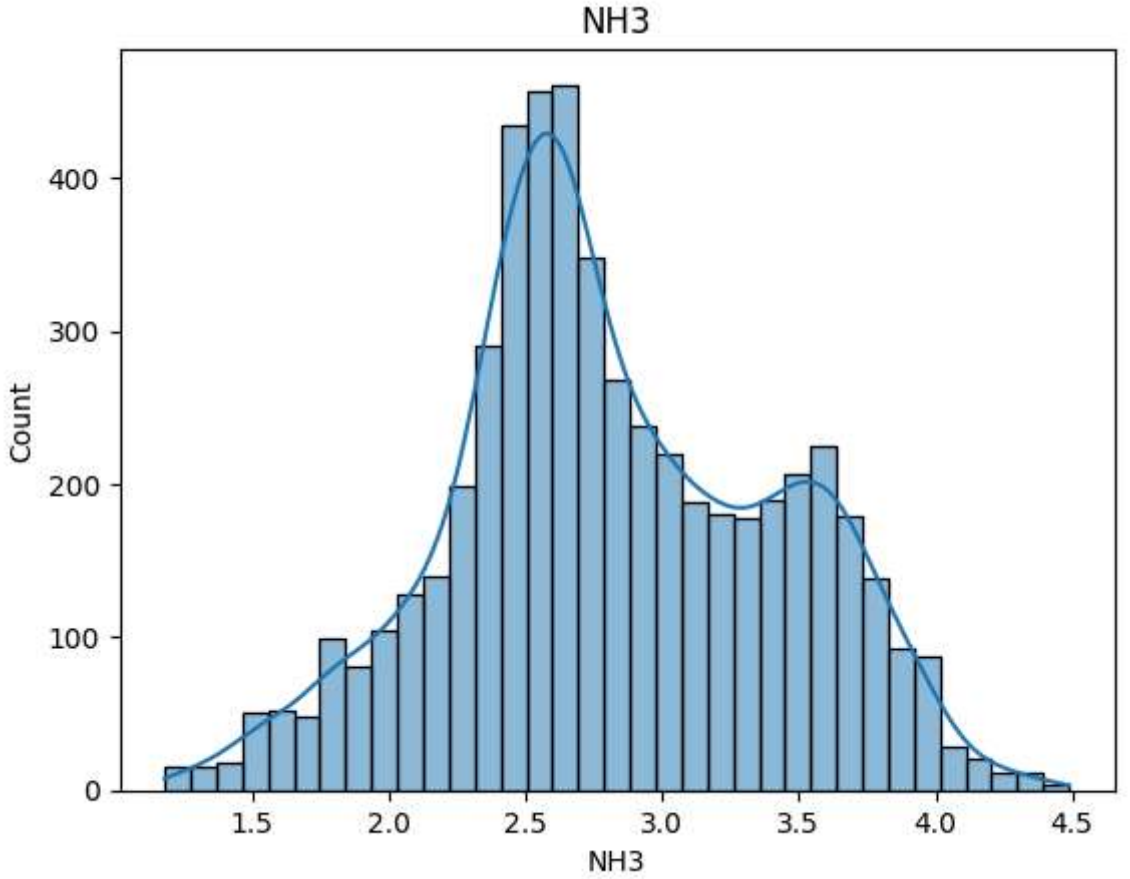

PM2.5
Count
0
50
100
150
200
250
300
350
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
PM2.5

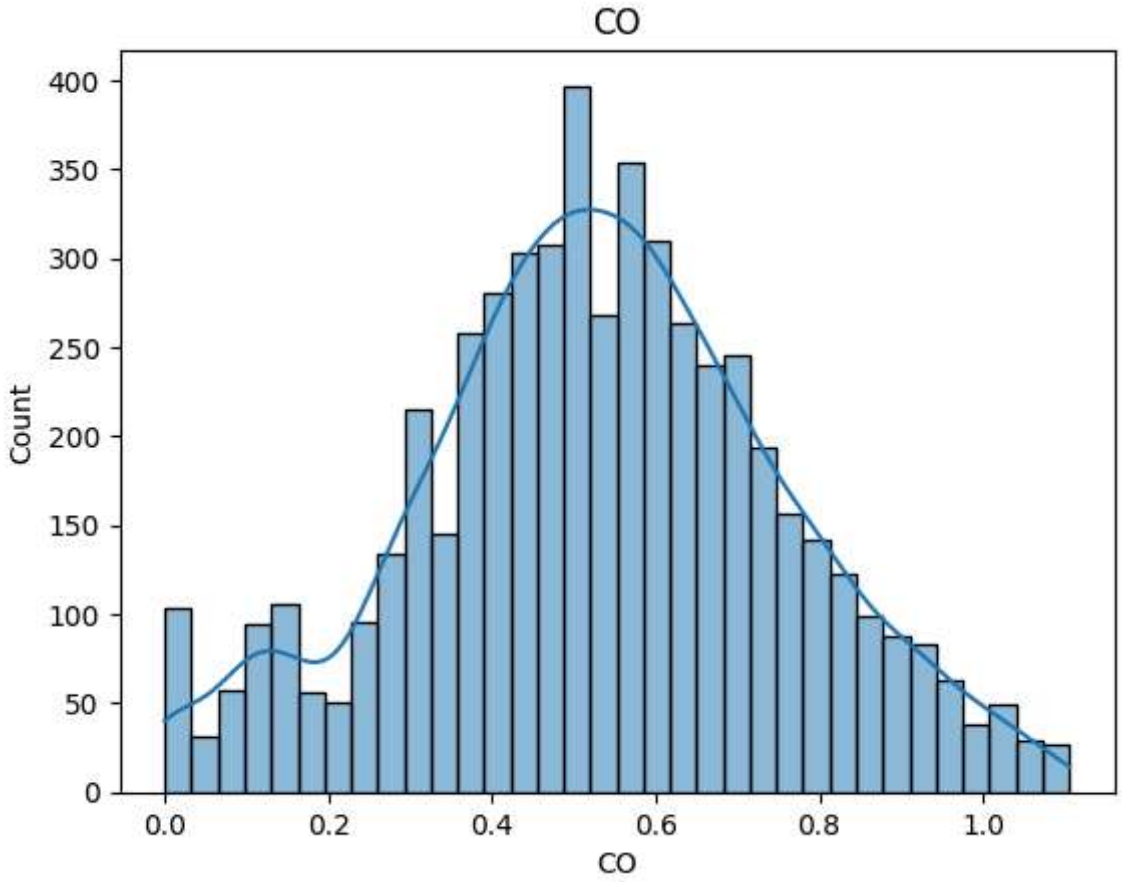

PM10
Count
0
50
100
150
200
250
300
350
400
3.0
3.5
4.0
4.5
5.0
5.5
6.0
PM10

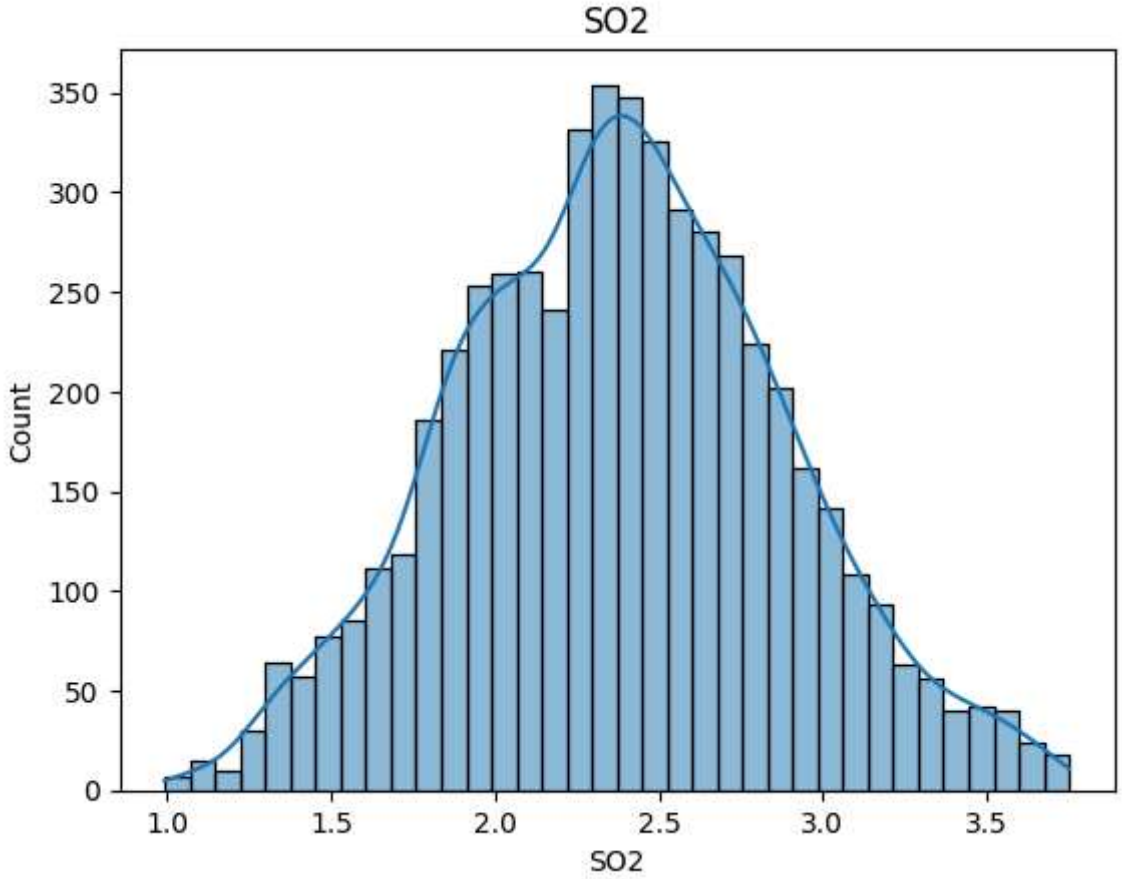

NO
Count
NO
0
50
100
150
200
250
300
350
1
2
3
4

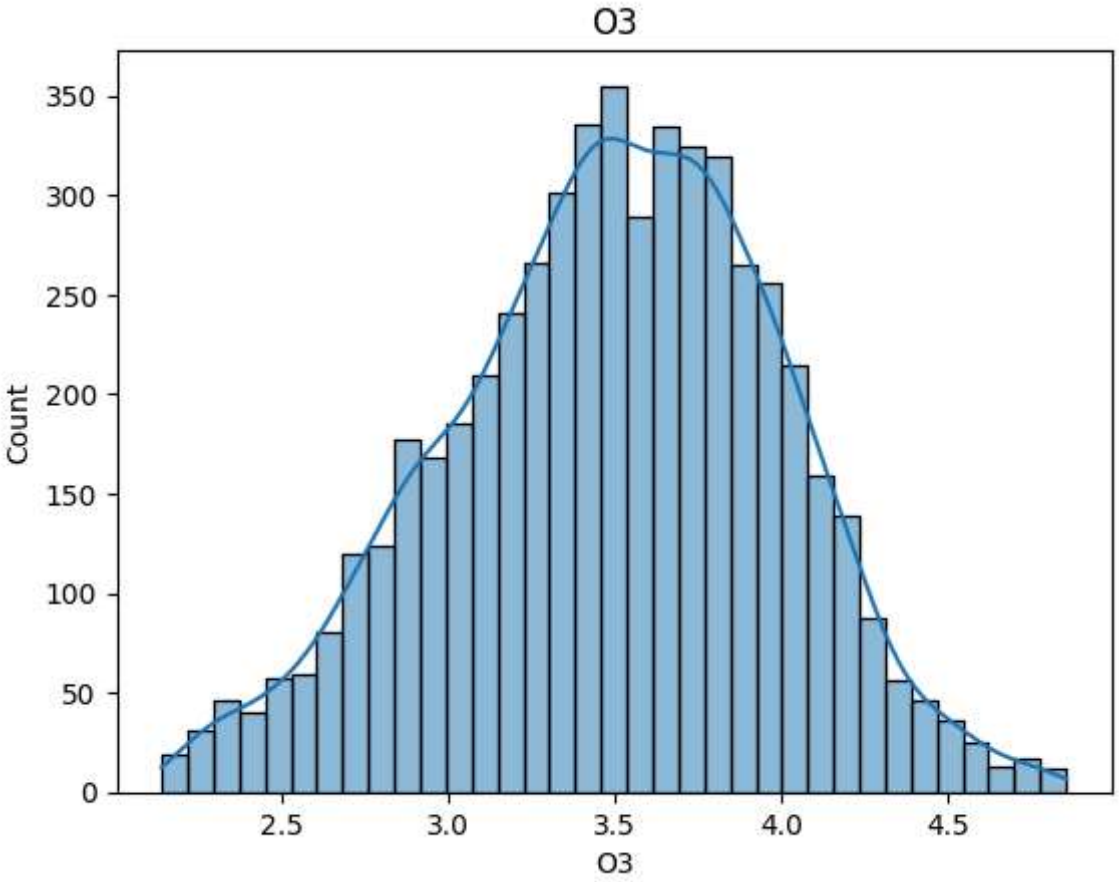

NO2
Count
NO2
0
50
100
150
200
250
300
350
400
1.5
2.0
2.5
3.0
3.5
4.0
4.5

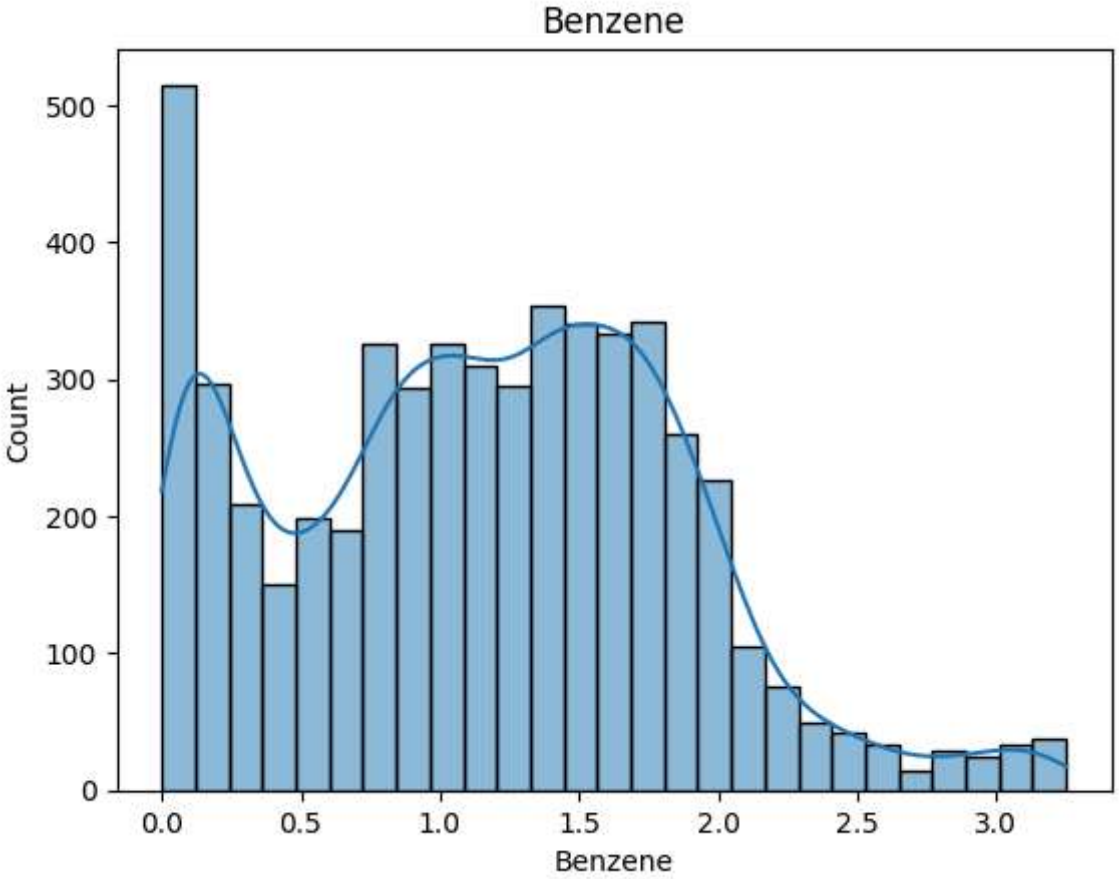

NOx
Count
NOx
0
50
100
150
200
250
300
350
400
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

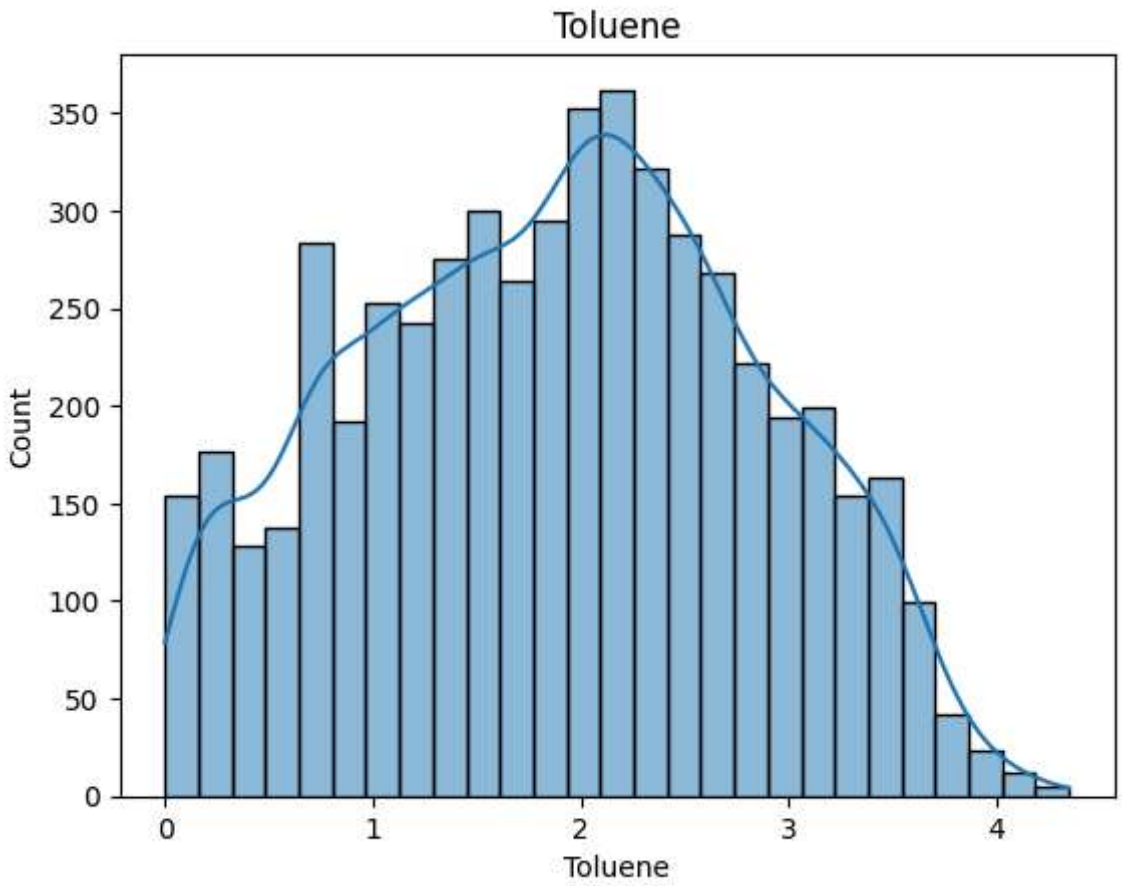

NH3
Count
NH3
0
100
200
300
400
1.5
2.0
2.5
3.0
3.5
4.0
4.5

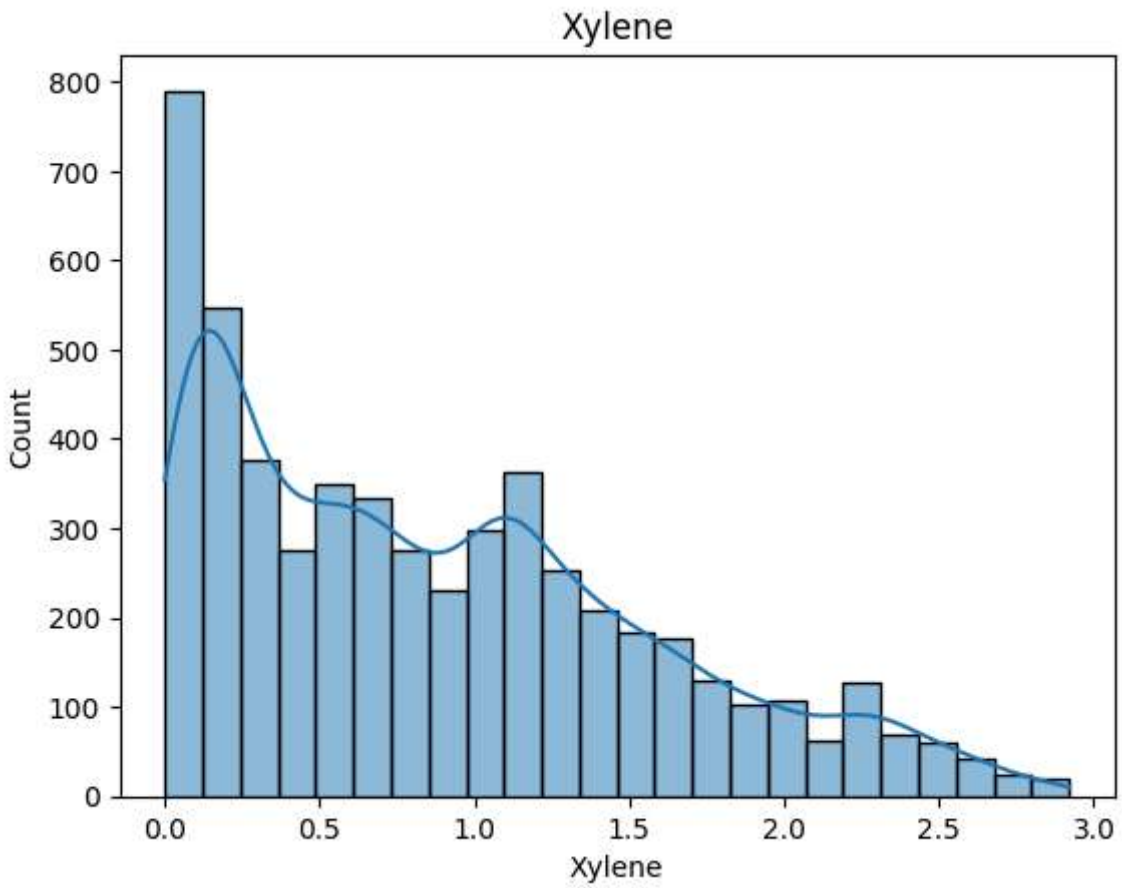

CO
Count
CO
0
50
100
150
200
250
300
350
400
0.0
0.2
0.4
0.6
0.8
1.0

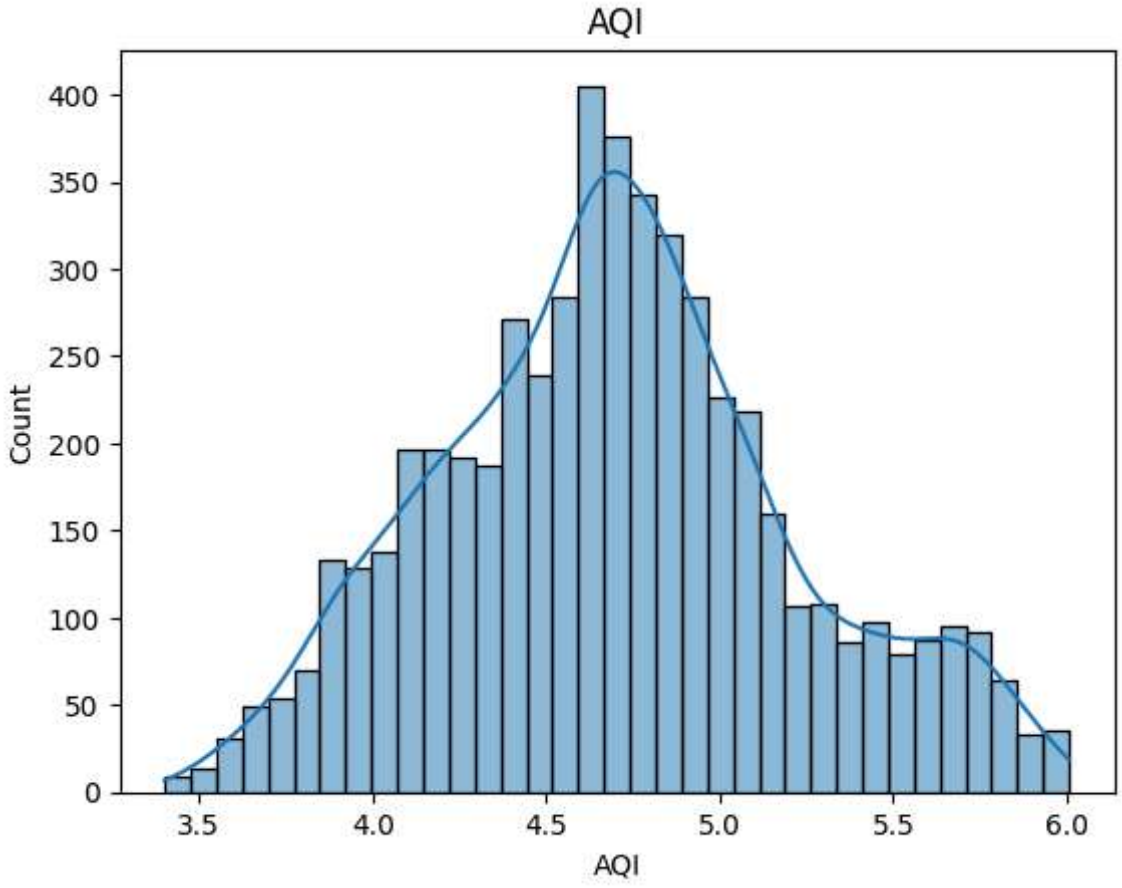

SO2
Count
SO2
0
50
100
150
200
250
300
350
1.0
1.5
2.0
2.5
3.0
3.5

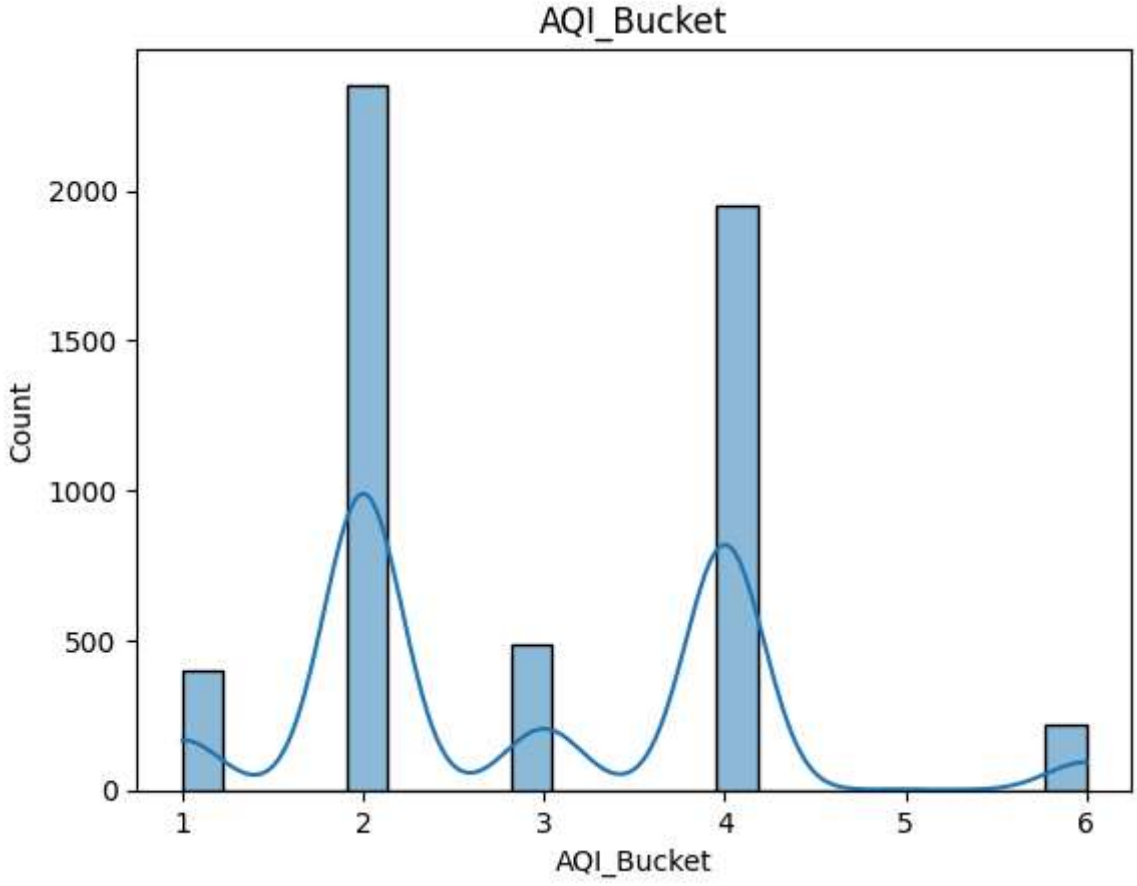

O3
Count
O3
0
50
100
150
200
250
300
350
2.5
3.0
3.5
4.0
4.5


Benzene
Count
Benzene
0
100
200
300
400
500
0.0
0.5
1.0
1.5
2.0
2.5
3.0


Toluene
Count
Toluene
0
50
100
150
200
250
300
350
0
1
2
3
4


Xylene
Count
Xylene
0
100
200
300
400
500
600
700
800
0.0
0.5
1.0
1.5
2.0
2.5
3.0

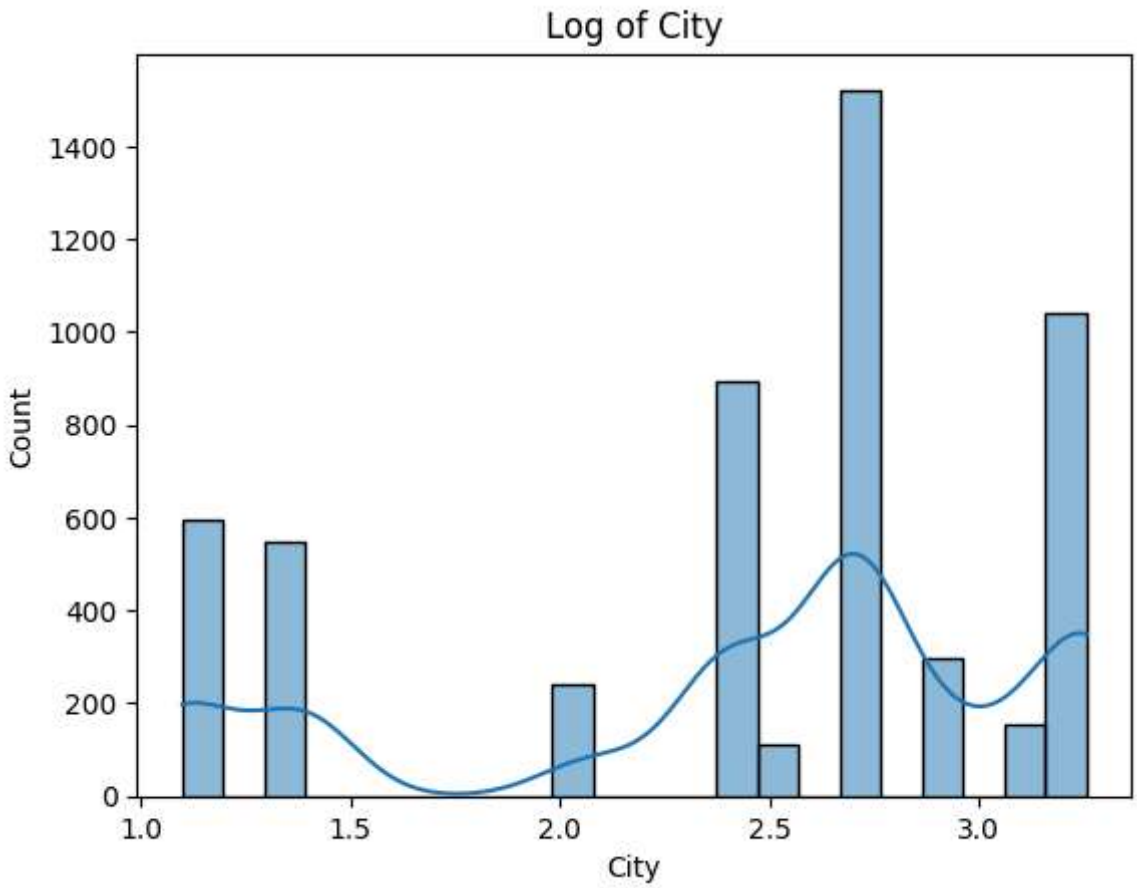

AQI
Count
AQI
0
50
100
150
200
250
300
350
400
3.5
4.0
4.5
5.0
5.5
6.0

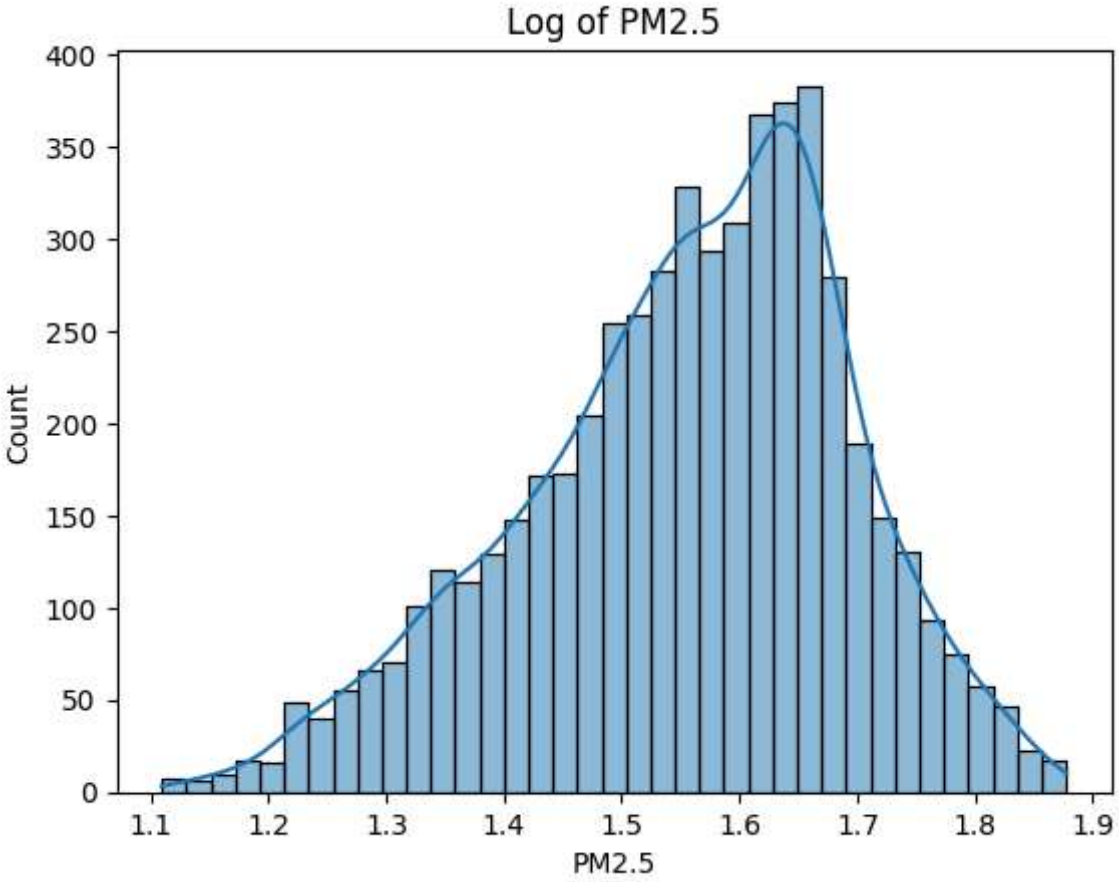

AQI_Bucket
Count
AQI_Bucket
0
500
1000
1500
2000
1
2
3
4
5
6

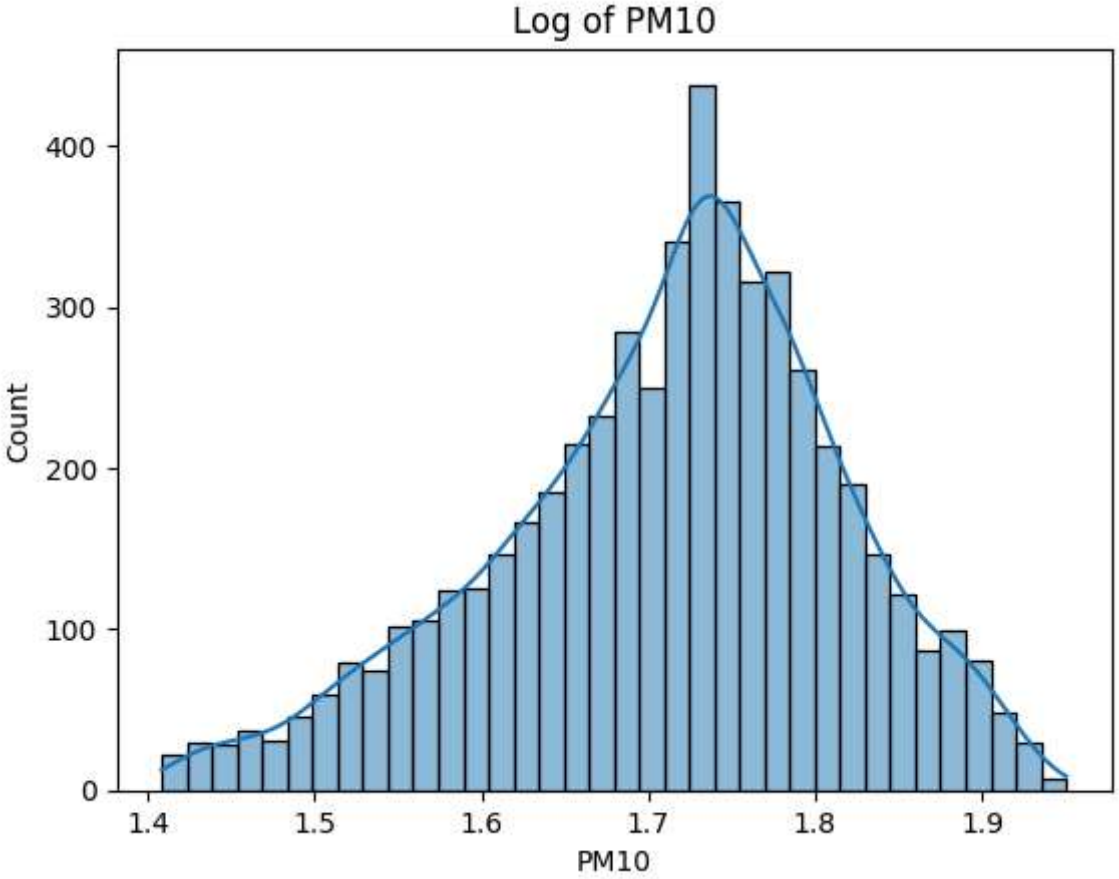
In [53]:
```
#for right skewed data
for col in df.columns:
    if df[col].skew() > 1:
        df[col] = np.log1p(df[col])   # log(1+x)
```

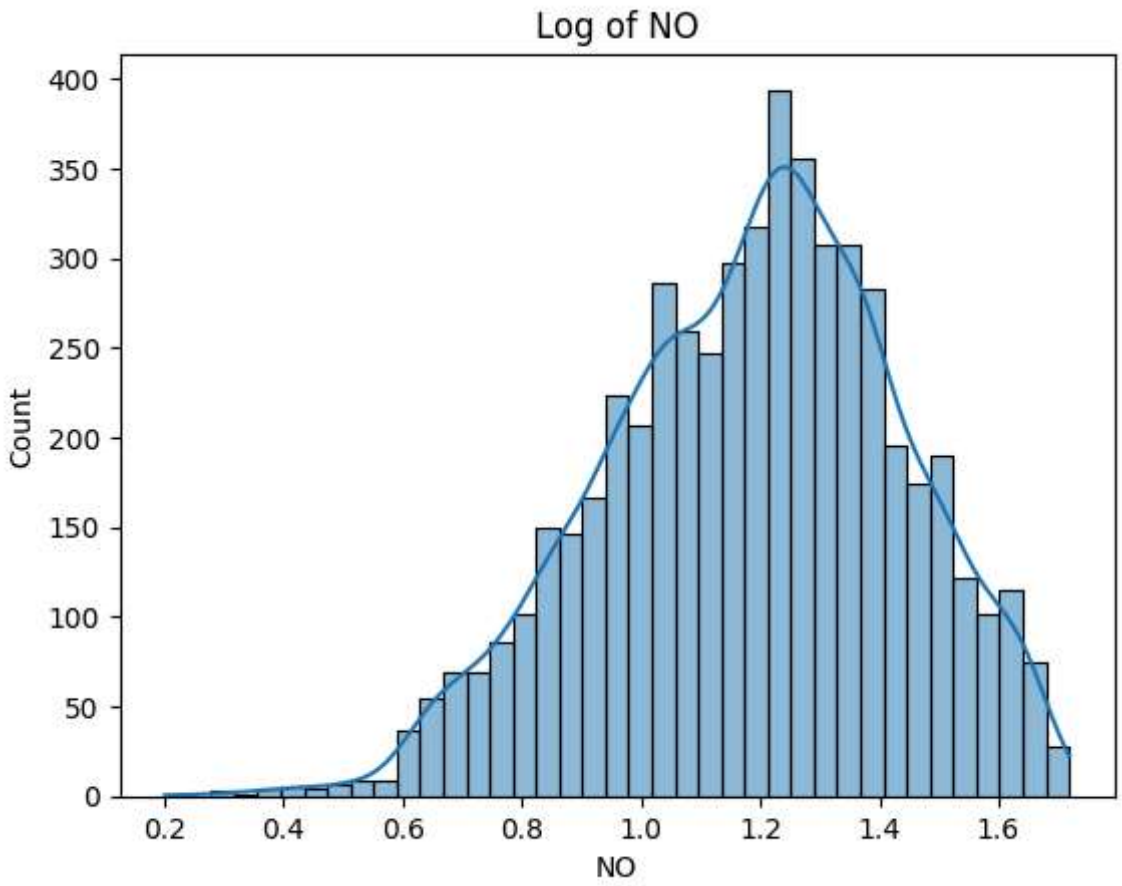
In [54]:
```
#for left skewed data
for col in df.columns:
    if df[col].skew() < -1:
        df[col] = np.square(df[col])
```

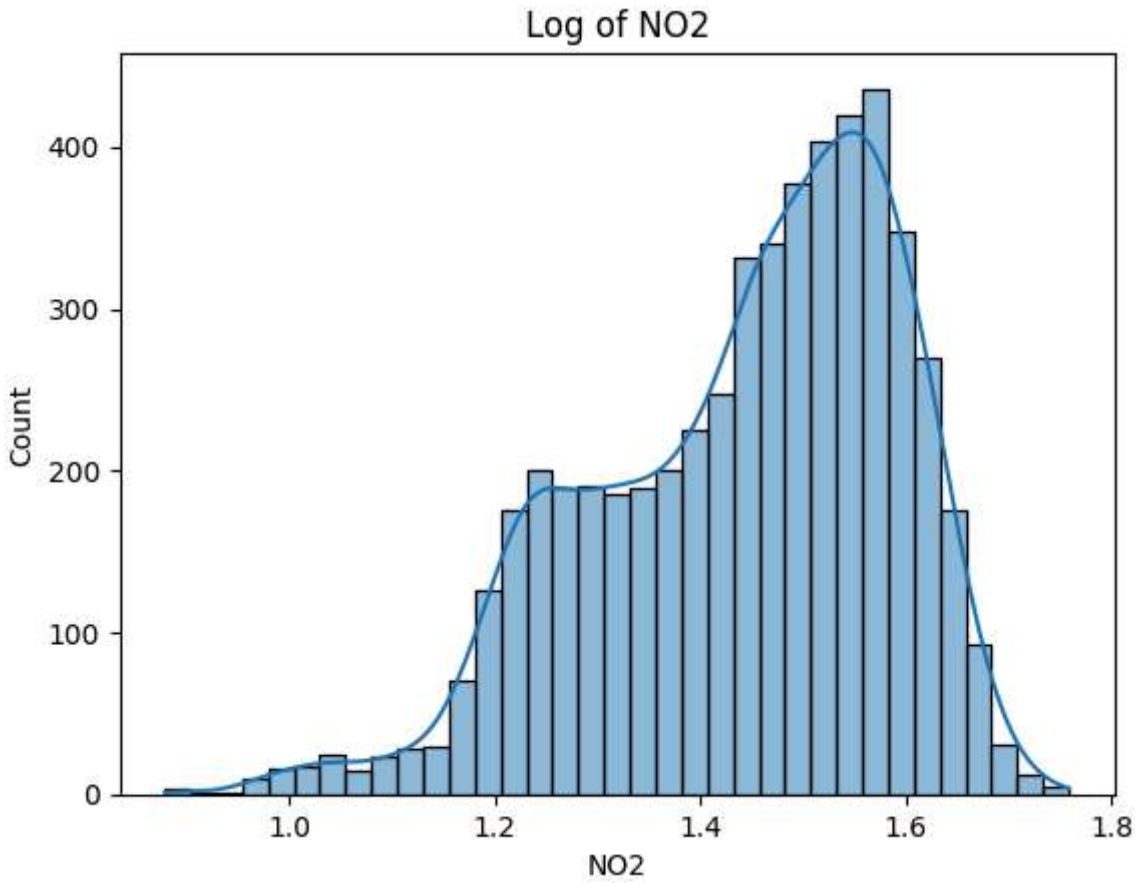
In [55]:
```
for col in df.columns:
    plt.figure()
    sns.histplot(np.log1p(df[col]), kde=True)
    plt.title("Log of " + col)
    plt.show()
```

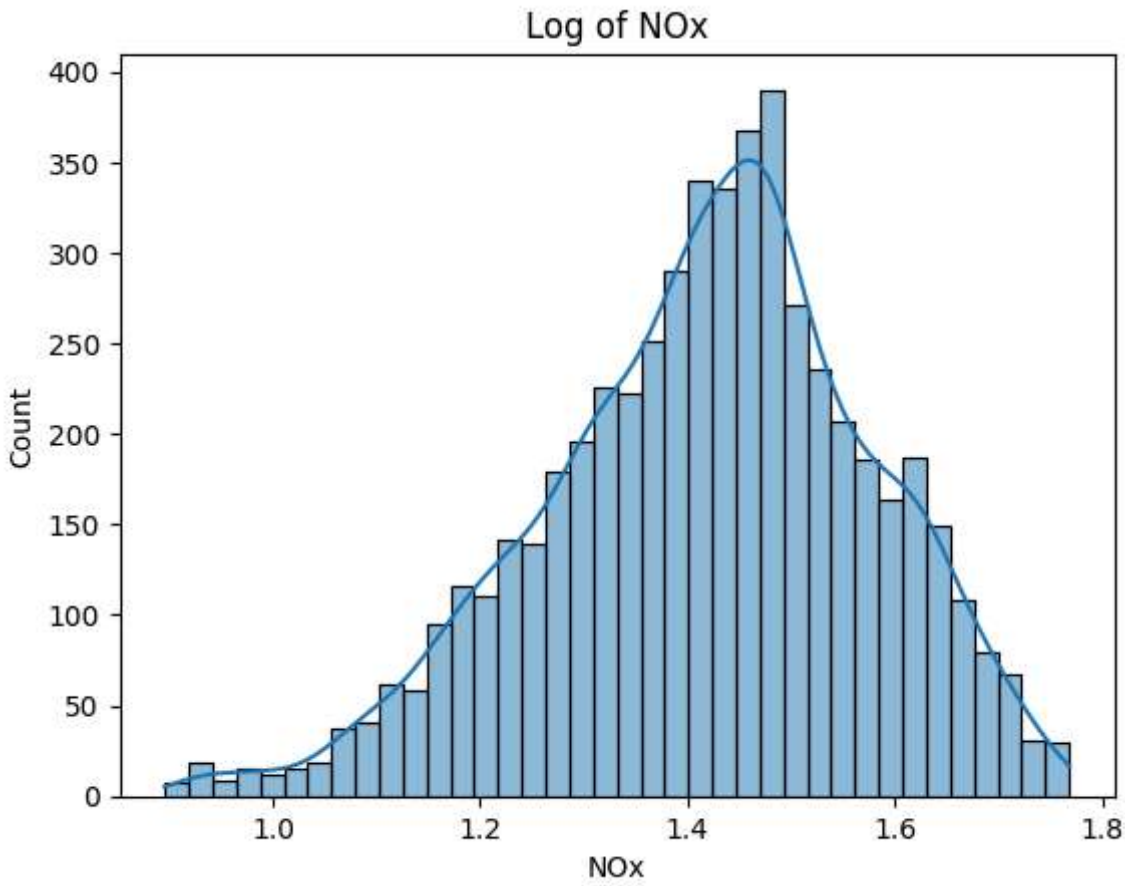


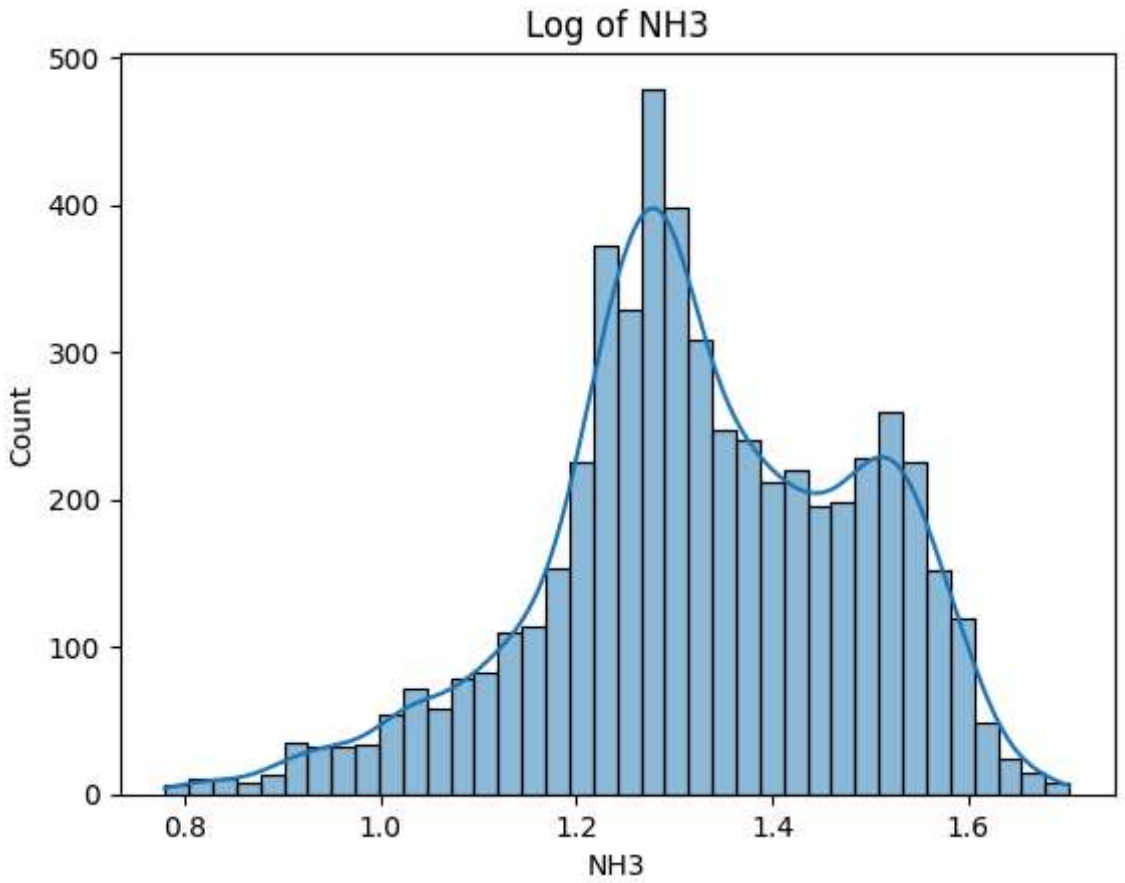

Log of PM2.5
Count
PM2.5
0
50
100
150
200
250
300
350
400
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9

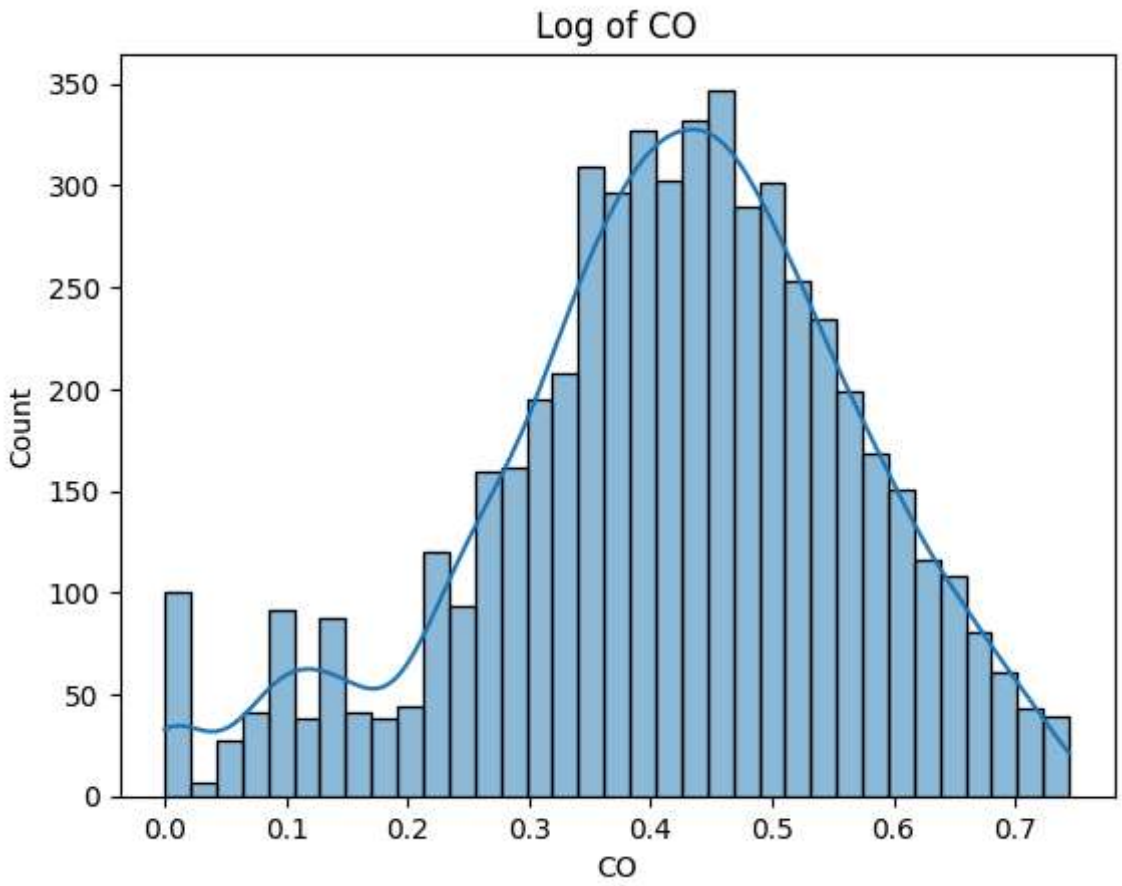

Log of PM10
Count
PM10
0
100
200
300
400
1.4
1.5
1.6
1.7
1.8
1.9

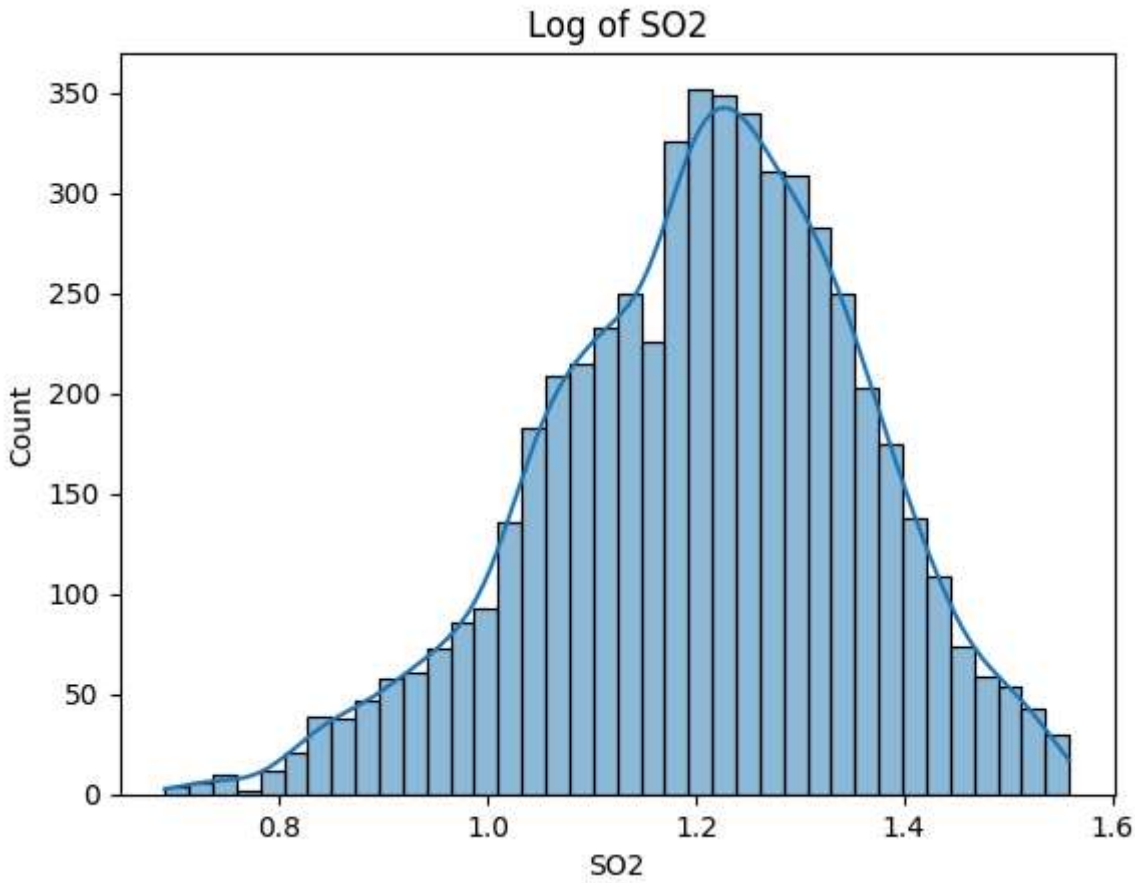

Log of NO
Count
NO
0
50
100
150
200
250
300
350
400
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6

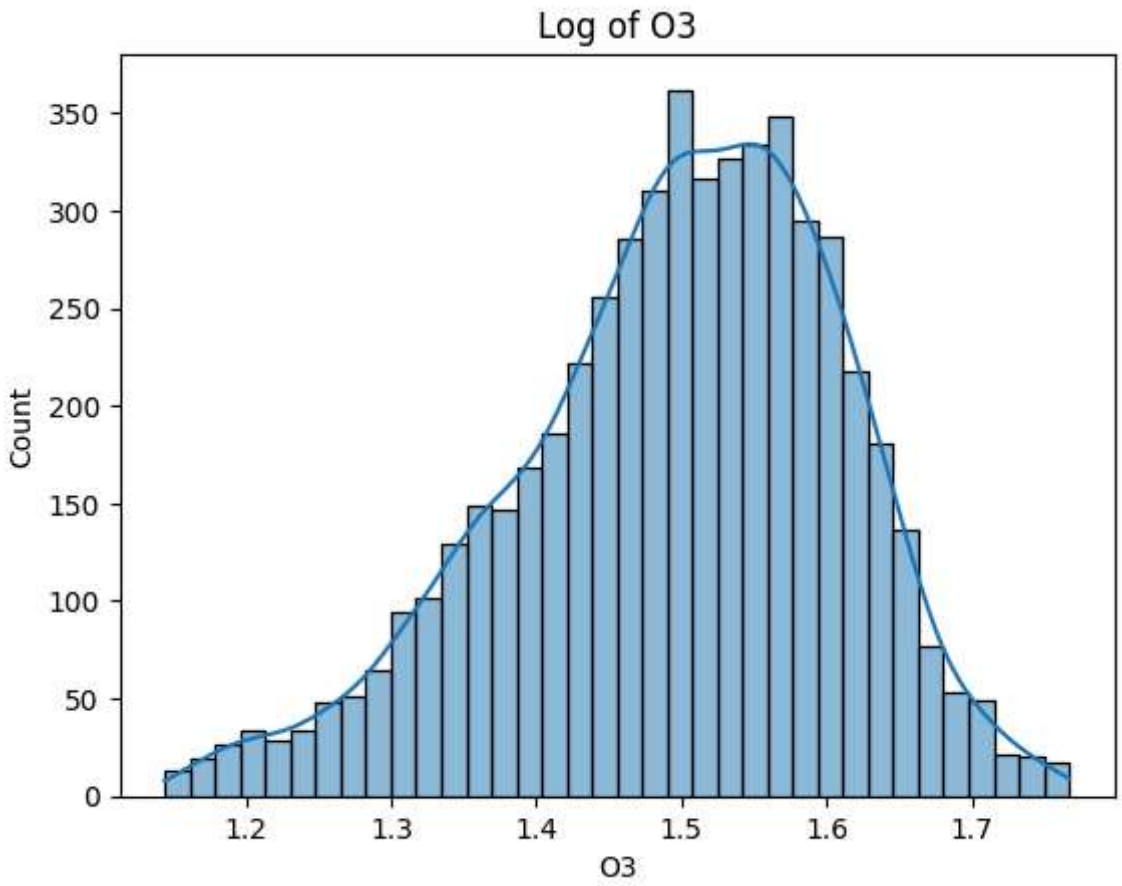

Log of NO2
Count
NO2
0
100
200
300
400
1.0
1.2
1.4
1.6
1.8

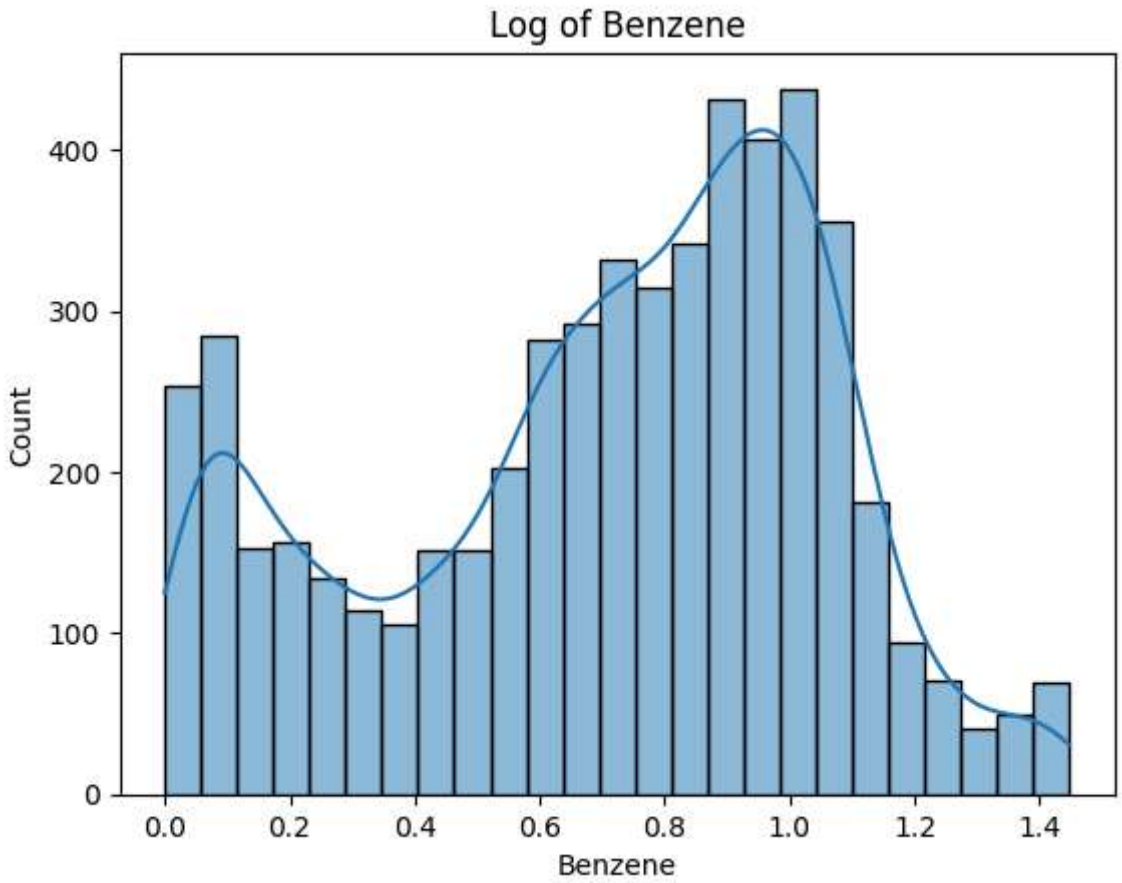

Log of NOx
Count
NOx
0
50
100
150
200
250
300
350
400
1.0
1.2
1.4
1.6
1.8

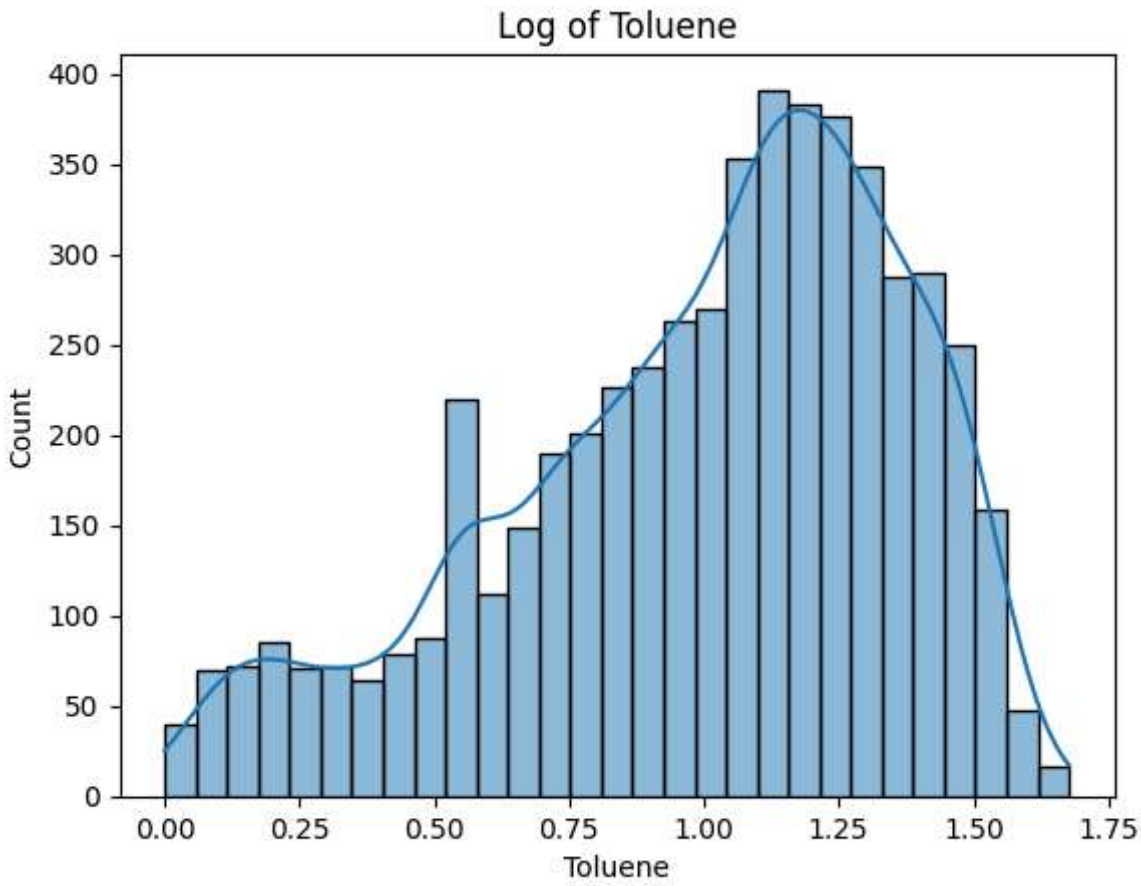

Log of NH3
Count
NH3
0
100
200
300
400
500
0.8
1.0
1.2
1.4
1.6

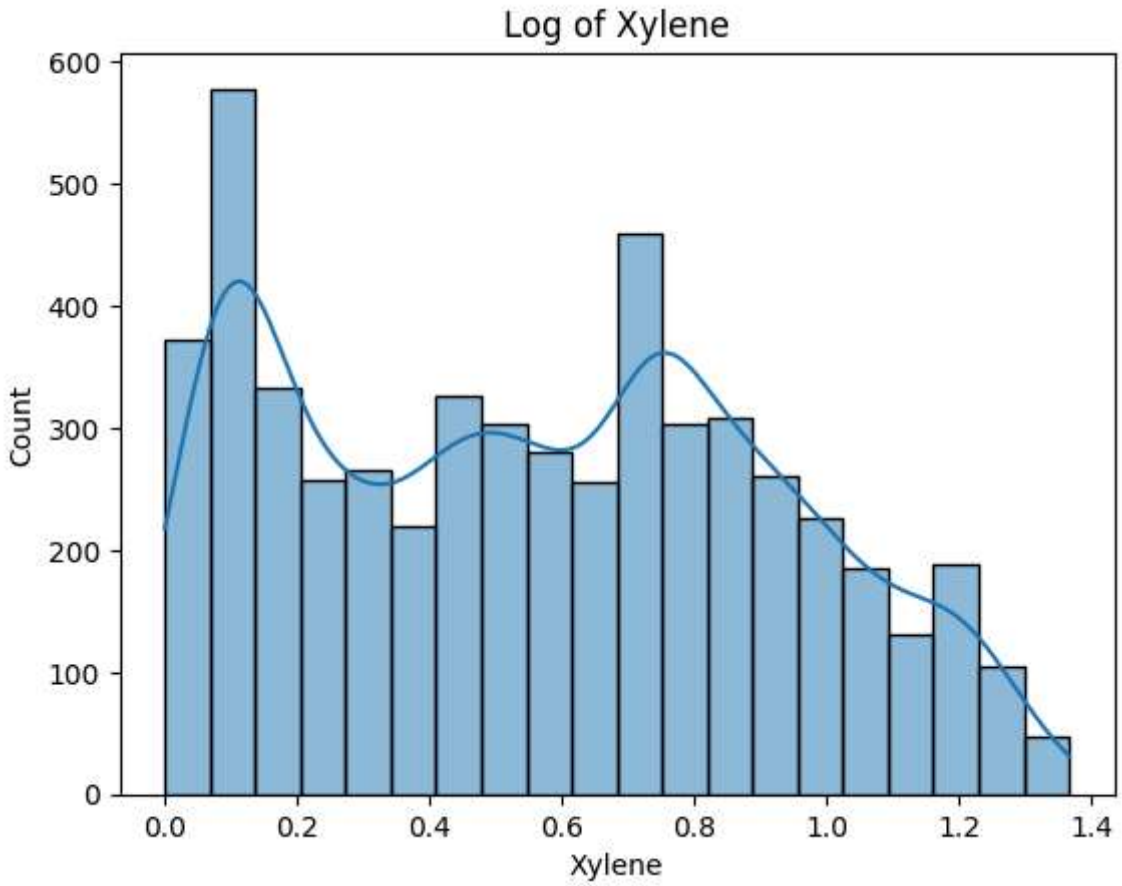

Log of CO
Count
CO
0
50
100
150
200
250
300
350
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7

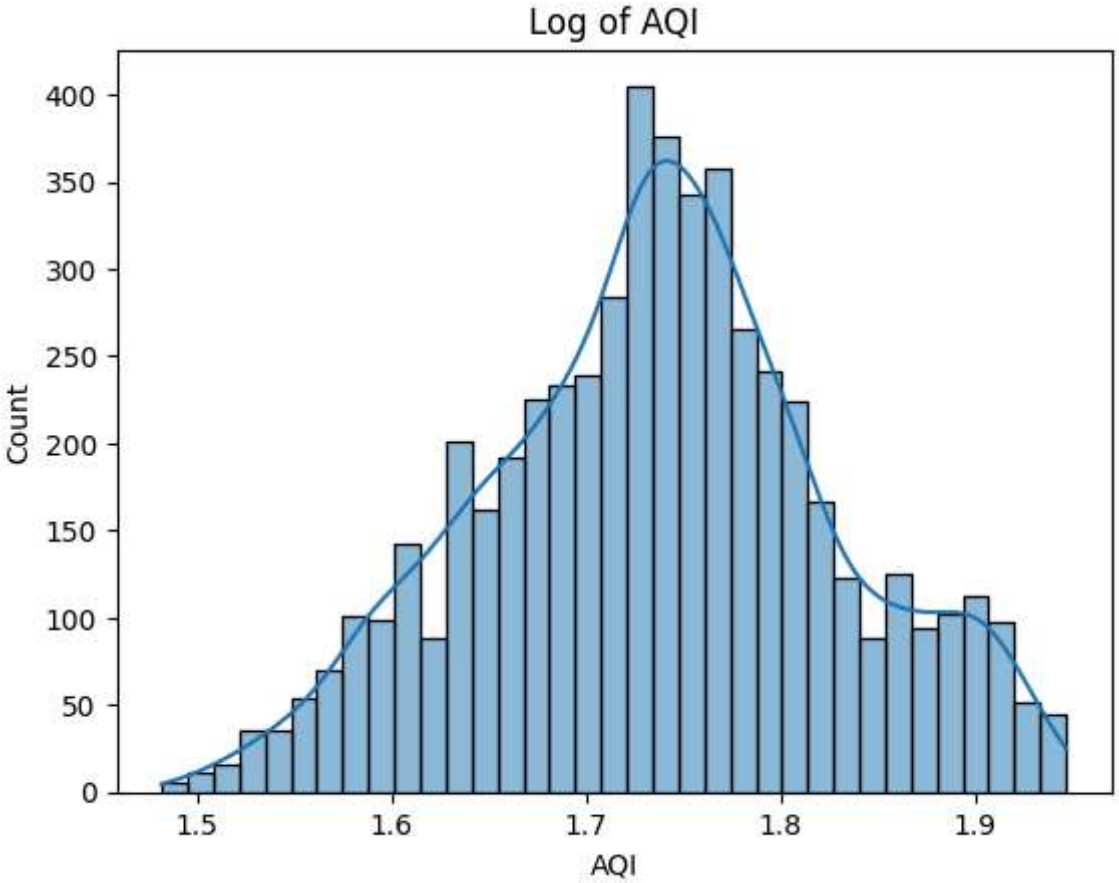

Log of SO2
Count
SO2
0
50
100
150
200
250
300
350
0.8
1.0
1.2
1.4
1.6

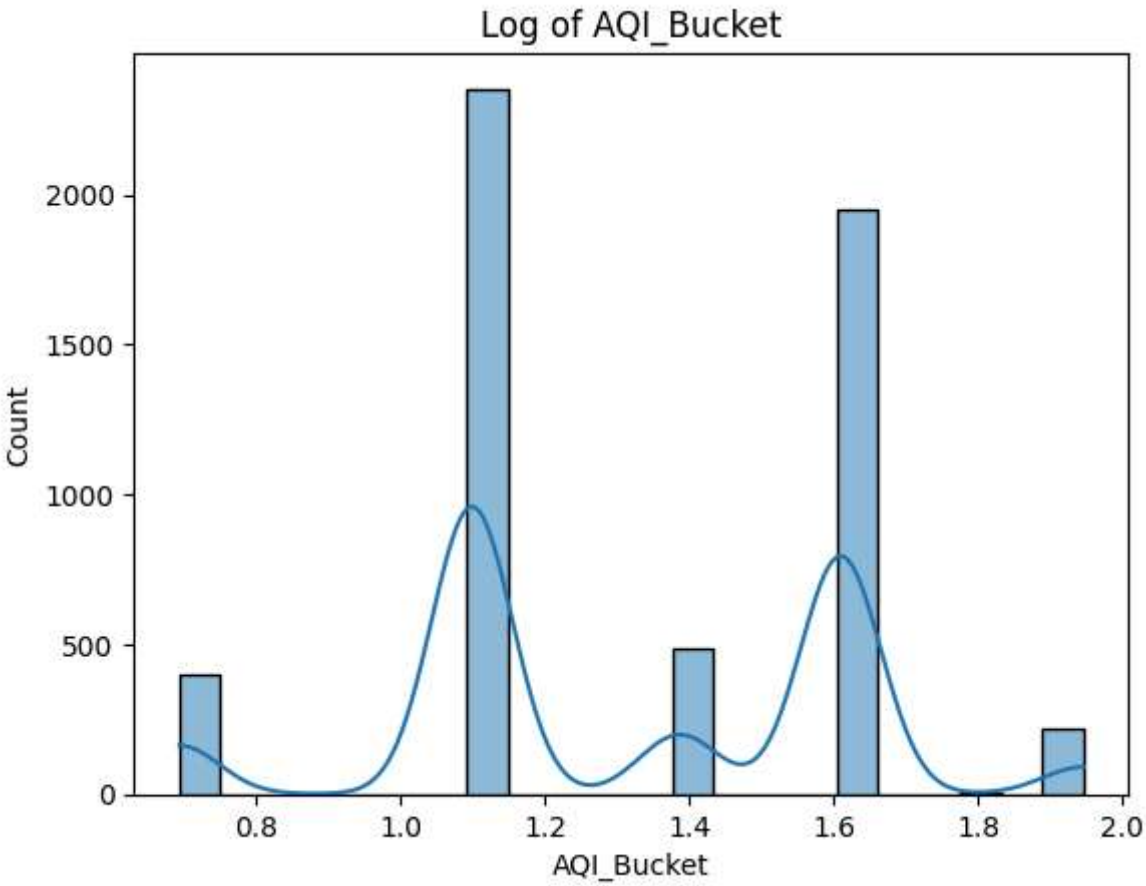

Log of O3
Count
O3
0
50
100
150
200
250
300
350
1.2
1.3
1.4
1.5
1.6
1.7


Log of Benzene
Count
Benzene
0
100
200
300
400
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

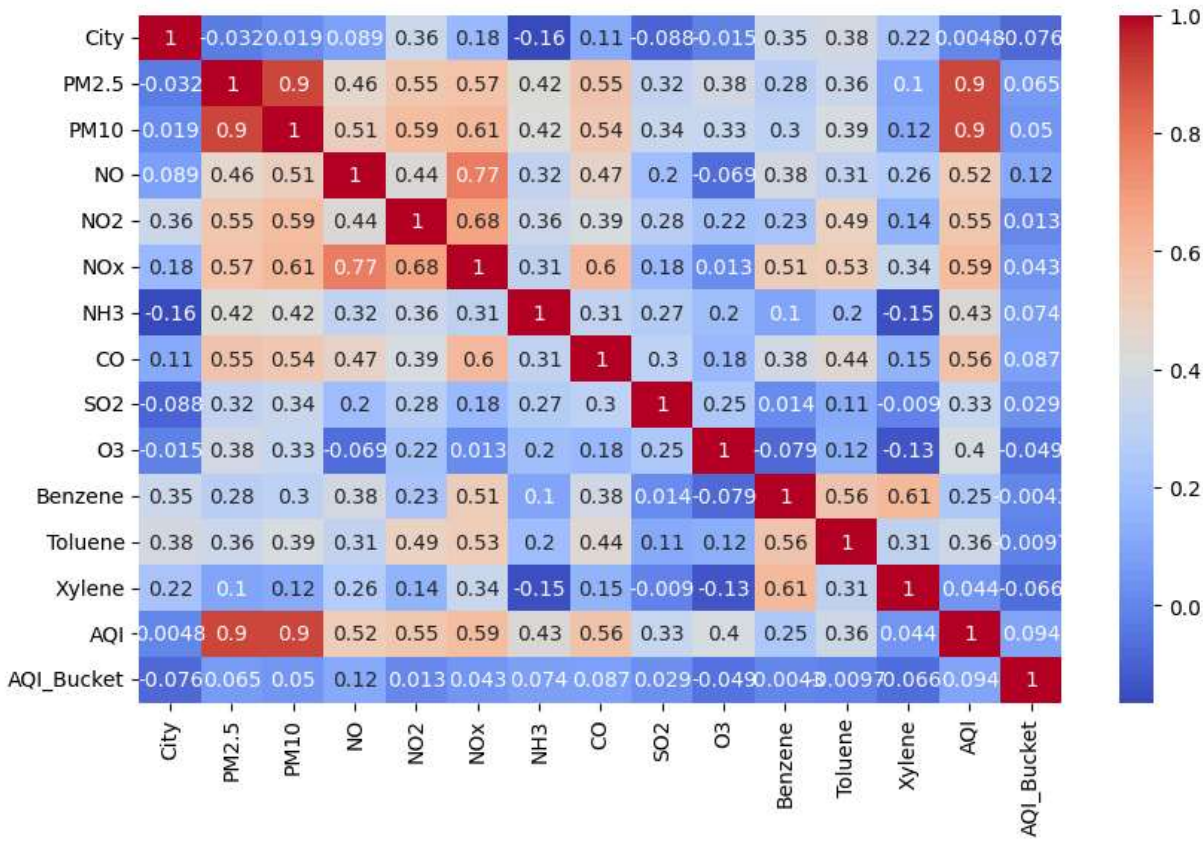

Log of Toluene
Count
Toluene
0
50
100
150
200
250
300
350
400
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75


Log of Xylene
Count
Xylene
0
100
200
300
400
500
600
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4


Log of AQI
Count
AQI
0
50
100
150
200
250
300
350
400
1.5
1.6
1.7
1.8
1.9


Log of AQI_Bucket
Count
AQI_Bucket
0
500
1000
1500
2000
0.8
1.0
1.2
1.4
1.6
1.8
2.0

In [56]:
```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



In [57]:
```
quantiles = df.quantile([0.25, 0.5, 0.75, 1.0])
print(quantiles.T)
```

```
                 0.25       0.50       0.75       1.00
City        7.000000  14.000000  18.000000  25.000000
PM2.5       3.342155   3.813748   4.223544   5.535640
PM10        4.199380   4.614922   4.939784   6.031238
NO          1.767296   2.347558   2.902520   4.574299
NO2         2.809704   3.356897   3.753144   4.807131
NOx         2.730464   3.182212   3.586154   4.859425
NH3         2.432297   2.721295   3.268808   4.488074
CO          0.392042   0.530628   0.678034   1.105257
SO2         2.023532   2.376764   2.714695   3.750915
O3          3.157638   3.510948   3.839775   4.852733
Benzene     0.590557   1.156881   1.656321   3.251149
Toluene     1.137833   1.934416   2.594508   4.349633
Xylene      0.246860   0.741937   1.316408   2.920470
AQI         4.343805   4.682131   5.010635   6.006353
AQI_Bucket  2.000000   2.000000   4.000000   6.000000
```